

Five OEIS Asymptotic Conjectures for Distinct-Partition Norms

Proofs, all-orders logarithmic expansions,
exact-saddle corrections, and inverse
and extreme-value laws

Research report prepared for Vladimir Reshetnikov
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Merged with an independent manuscript on A022629
(batch 73O1: base source 40, member source 48)

Central family

$$A_\alpha(q) = \prod_{k \geq 1} (1 + k^\alpha q^k), \quad a_\alpha(n) = [q^n] A_\alpha(q), \quad \alpha > 0.$$

The cases $\alpha = 1, 2, 3, 4, 5$ are [A022629](#), [A092484](#), [A265840](#), [A265841](#), and [A265842](#). All five consulted entries label their leading logarithmic asymptotic as a conjecture. This report proves those formulas and develops a single refinement mechanism for every fixed positive α .

Scope and verification. The results below have detailed English proofs and accompanying exact and numerical checks. They are not claimed to be Lean-verified or peer-reviewed. The literature comparison is targeted, not an exhaustive priority certification. The complete exponentially improved coefficient transseries is not claimed: the report distinguishes its proved logarithmic block, exact-saddle expansion, and Fourier-resonance scale. Source 48, merged here, states the same scope for $\alpha = 1$ and real $\alpha > 0$: no Lean verification, no worldwide priority, and no convergence of the inverse-logarithmic series.

Abstract

For a partition λ into distinct parts, let $N(\lambda)$ be the product of its parts. We analyze the fixed positive norm moments $a_\alpha(n) = \sum_{\lambda \vdash n}^{\text{distinct}} N(\lambda)^\alpha$. A direct argument proves the logarithmic growth formulas still marked conjectural in five OEIS entries. With $K = \sqrt{2n}$, $r = \log K$, and $c = \pi^2/(6\alpha^2)$, we then prove, to every fixed order,

$$\log a_\alpha(n) = \alpha K \left(r - 1 + \frac{c}{r} + \frac{c}{r^2} + \frac{c - c^2/2}{r^3} + \dots \right).$$

An explicit inverse-function construction generates all coefficients; seven are printed. A separate local-limit argument gives multiplicatively accurate exact-saddle asymptotics and all fixed Edgeworth orders. This separation matters: exponentiating a fixed truncation of the logarithmic expansion does not give a relative asymptotic equivalent. We invert the dominant block using the principal Lambert function and obtain asymptotics for the first index attaining a large value. Under the norm-biased measure, the occupation boundary has a logistic profile, while the largest part has a Gumbel limit and is asymptotic to $(1 + 1/\alpha)\sqrt{2n}$. Finally, we prove that the first Fourier resonances have logarithmic modulus of order $-K/(\alpha^3(\log K)^3)$, explaining a striking finite-size limitation visible in the exact computations. Reproducible scripts, source-status notes, and further research questions accompany the proofs.

This report merges two manuscripts that proved these theorems independently: source 40 (the base text, five OEIS sequences, real $\alpha > 0$) and source 48 (A022629 and the same real family, written $\prod(1 + k^s q^k)$). Every shared theorem is printed once and credited to both, with identical coefficients. Source 48 adds a second, elementary proof of the upper bound, a second derivation of all coefficients by an explicit differential operator, explicit centring of the Gumbel law in $\log \sqrt{2n}$, the radius of convergence of $\log A_\alpha$ at the origin, the logarithm of the largest single weight, the entropy factor $\log(a_\alpha(n)/\max N^\alpha) \sim \pi^2 K/(6\alpha r)$, exact data to $n = 6400$ and its own research questions.

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1 The problem, its significance, and the source audit

1.1 Five conjectures in one natural family

Let $\mathcal{D}_n$ be the set of partitions of n into distinct positive parts. The empty partition has norm 1. For fixed $\alpha > 0$, define

$$N(\lambda) = \prod_{k \in \lambda} k, \quad a_\alpha(n) = \sum_{\lambda \in \mathcal{D}_n} N(\lambda)^\alpha. \quad (1.1)$$

Choosing whether each part occurs gives the locally uniformly convergent product

$$A_\alpha(q) = \sum_{n \geq 0} a_\alpha(n) q^n = \prod_{k \geq 1} (1 + k^\alpha q^k), \quad |q| < 1. \quad (1.2)$$

For integral α the coefficients are integers. One may interpret k^α as the number of colors available for a part of size k , with at most one part of each *size*, regardless of color. This is not the product $\prod_k (1 + q^k)^{k^\alpha}$, which allows several differently colored copies of the same size.

The OEIS entries consulted on October 1, 2026 have the following status [1, 2, 3, 4, 5].

α	Sequence	Date of the conjectural formula
1	A022629	May 8, 2018
2	A092484	December 27, 2020
3	A265840	December 27, 2020
4	A265841	December 27, 2020
5	A265842	December 27, 2020

All five formulas are attributed there to Václav Kotešovec and have the unified form

$$\log a_\alpha(n) \sim \alpha \sqrt{\frac{n}{2}} (\log(2n) - 2). \quad (1.3)$$

The full array of nonnegative integral moments is [A292189](#); its columns $0, 1, \dots, 5$ are identified in that entry [6]. The zeroth column is the ordinary distinct-partition function, but $\alpha = 0$ is a different asymptotic regime and is excluded from our fixed-positive-parameter theorems.

Source 48, which treats A022629 and the real family, makes two further points [1]. Its use of “currently” refers to the entry as retrieved on October 1, 2026, not to an exhaustive determination of the publication history. And a logarithmic equivalent does not determine a multiplicative equivalent: even the constant -2 inside the parentheses of (1.3) is not specified by the relation $\sim$, since changing it leaves the same equivalence. The additive statement that the corresponding $-\sqrt{2n}$ term is correct is sharper, and Proposition 3.2 proves it. For non-integral α , (1.2) is a positive weighted enumeration, and the product converges normally on compact subsets of $|q| < 1$ (source 48).

These sequences are not arbitrary numerical constructions. They simultaneously encode weighted enumeration, moments of a fundamental multiplicative partition statistic, and a conditioned model of independent binary occupancies. Their common structure makes a family theorem more informative than five separate fits to numerical data.

1.2 What is established here, and what is not a priority claim

The conjectural labels in the consulted entries are verifiable facts; the claim that no proof has ever appeared anywhere would be much stronger and is not made. This report supplies self-contained proofs of the displayed formulas, followed by explicit refinements and probabilistic

consequences. Novelty of the refinements should be assessed against the broader literature before publication.

Schneider and Sills survey partition norms, including extremal norms over distinct partitions and their connection with triangular numbers [7]. Bridges and Craig study norm distributions and moments for *unrestricted* partitions [8]; repeated parts change the analytic problem substantially. Rana, Kaur, and Kumar give recurrences and related norm identities for distinct partitions [9]. None of those results is used as an unproved asymptotic input here.

A particularly relevant recent source is Gikunda’s March 2026 dissertation [10]. Its Chapter 4 uses $\prod_k (1 + u^{\log k} q^k)^{b_k}$ and proves log-norm central limit theorems. For $b_k = 1$ and $u = e^\alpha$ this is our generating family. However, the coefficient estimate in Proposition 4.2.5 and the surrounding analysis use a *shrinking* tilt $\log u = O(t^\delta)$ as the saddle parameter t tends to zero. Our tilt $\alpha > 0$ stays fixed. Thus that estimate does not directly supply the fixed-tilt results below. This is a distinction between asymptotic regimes, not a claim that the underlying family has not previously been studied.

Source 48 places the method in the classical saddle-point and probabilistic tradition of asymptotic enumeration; standard background is in Flajolet and Sedgewick [13]. Granovsky and Stark [14] study a broad product framework and, in their Section 3, the related products $\prod (1 + k^{-q} z^k)$ with $q > 0$. Their main theorem assumes parameters $0 < a_k \leq 1$, so substituting $a_k = k^\alpha > 1$ is not a direct application of that theorem. This observation is not a claim that no other general theorem can cover the present case (source 48).

1.3 Relation to the ProveIt repository

The repository snapshot consulted was `1f1981f682b2878bde51a6ad40c22777f362fc05`. Its `Analysis/Transseries/README.md` describes the transseries and inversion volumes, accompanying verification programs, and the separate Lean library for asymptotic scales and power-logarithmic monomials [11]. The present report follows that program by separating three precision levels: a logarithmic asymptotic block, a relative exact-saddle expansion, and flat Fourier contributions. Its Lambert- W inversion is given explicitly rather than left as a formal instruction to “reverse the series.”

No repository theorem is assumed here without proof. In particular, the existence of general transseries infrastructure is not treated as a formal certificate for our analytic remainder estimates. The new material is an independent, reproducible research contribution suitable for integration after review; it does not modify the repository.

Source 48 read the repository’s `Analysis/Transseries` material, which consolidates polynomial-logarithmic asymptotics, Lambert-core inversion, error transport, and staircase inversion [12]; those are the organizing ideas it uses. No repository theorem is imported as an unexamined analytic hypothesis: the needed estimates are proved below. A targeted repository search for A022629 returned no match during its preparation. That search and the literature checks are a useful duplication screen, not a proof of originality (source 48).

[Merge note, batch 73O1.] The repository still contains no other treatment of these sequences; this report is the first. Its methodological relative is the partition-number chapter of the volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion`, which treats $p(n)$ (A000041); nothing is imported from it.

1.4 The two sources and this merge

This report is built from two manuscripts of batch 7301 of ProveIt’s incoming-reports intake, written independently on October 1, 2026.

- **Source 40** (the base text), *Five OEIS Asymptotic Conjectures for Distinct-Partition Norms*: the family $\prod(1 + k^\alpha q^k)$ for real $\alpha > 0$, the five OEIS sequences $\alpha = 1, \dots, 5$, all fixed orders with seven printed coefficients and an eighth checked numerically, the exact-saddle expansion, Lambert- W inversion, the logistic boundary and Gumbel maximum, the fixed-resonance theorem, and finite-size diagnostics to $n = 5000$. Its section order is kept.
- **Source 48**, *The A022629 Conjecture: All-Orders Logarithmic Asymptotics, Saddle-Point Expansions, Inverse Growth, and the Largest Part of a Weighted Partition*: the same theorems for A022629 and for $\prod(1 + k^s q^k)$, real $s > 0$, with six printed coefficients. Its s is source 40’s α ; with that identification its coefficients $c_j(A)$, $A = \pi^2/(6s^2)$, are exactly source 40’s $P_j(c)$, $c = \pi^2/(6\alpha^2)$, and its inverse coefficients and first Edgeworth correction are identical.

Every result, proof, remark, question and non-claim of both manuscripts is printed. A theorem proved by both is printed once, in source 40’s wording, and credited “sources 40 and 48”. Where source 48’s proof is genuinely different, it follows as a “Second proof (source 48)”. Material only source 48 has is marked (source 48): Propositions 3.4 and 3.5, Lemma 3.6, Corollary 8.1, the operator route of Section 7.3, the explicit forms in Sections 9 and 10.3, the computations of Section 12.5, its questions, its OEIS wording, and Appendix C. Text added in the merge is marked [Merge note, batch 7301.]. Appendix D records the provenance and every choice the merge made.

1.5 Notation across the two sources

The two manuscripts use the same letters for different objects. This report uses source 40’s notation throughout; source 48’s material is translated as follows.

Notion	This report (source 40)	Source 48
power in the weight	α	s (renamed)
generating function, coefficients	$A_\alpha(q)$, $a_\alpha(n)$	$F_s(q)$, $a_s(n)$; $F(q)$, $a(n)$ for A022629
$\sqrt{2n}$, $\log \sqrt{2n}$	K , r	x , λ
$\pi^2/(6\alpha^2)$; coefficients	c ; $P_j(c)$	A ; $c_j(A)$
log of the transition site	$s = -W_{-1}(-t/\alpha)$	L
transition site e^s	M	r
$\log A_\alpha(e^{-t})$	$F(t)$	$S_s(t)$
mean, variance	$-F'(t)$, $B = F''(t)$	$m(t)$, $V(t)$
Bernoulli variables, total	X_k , $S = \sum kX_k$	B_k , X_t
log-expansion series	$\mathcal{H}_\alpha(r)$	$\mathcal{H}_s(\lambda)$
boundary operator and series	$\mathcal{T}$, $\mathcal{C}_\alpha$, $\mathcal{B}_\alpha$	$\mathcal{T}$, $\mathcal{C}_s$, $\mathcal{B}_s$
truncation order (log, Edgeworth)	R , J	K , M
block length (Lemma 5.1)	h	H
largest single weight	$\Pi_\alpha(n)$	$Q_s(n)$
largest part	L_n	M_n
log threshold; Lambert data	Y ; w , R_0 , K_0	y ; w , ℓ , x_0 (and $Y = y/s$)
explicit Gumbel centring	$\beta_\alpha(r)$	$d_s(\lambda)$
small exponent in proofs	σ , β	α

Readings to avoid.

- Source 48's s is the *power* ($= \alpha$); source 40's s , used here, is the *logarithmic transition coordinate* $-W_{-1}(-t/\alpha)$, which source 48 calls L .
- Source 48's $r = e^L$ is the transition *site* (here M); source 40's r , used here, is $\log \sqrt{2n}$ (source 48's λ).
- Source 48's K is a truncation order or a cutoff index; here K is always $\sqrt{2n}$. Source 48's M is an Edgeworth order; here M is always e^s .
- Source 48's $Q_s(n)$ is the largest single weight (here $\Pi_\alpha(n)$), not source 40's mean-relation series $Q_\alpha(s)$. Source 48's Gumbel centring $d_s(\lambda)$ (here $\beta_\alpha(r)$) is not source 40's boundary series $d_\alpha(s)$.
- B is the variance; $\mathcal{B}_\alpha$ is source 48's mean-boundary series.
- Source 48's Edgeworth sum $\mathcal{E}_M$ is cumulative (all weights below $2M$); source 40's $\mathcal{E}_\ell$ is the single weight 2ℓ . Here $1 + \sum_{\ell < J} \mathcal{E}_\ell$ is source 48's $\mathcal{E}_J$.

2 Main results and the hierarchy of scales

Throughout the paper, $\alpha > 0$ is fixed. Constants in O -terms may depend on α and on the indicated truncation order. Put

$$K = \sqrt{2n}, \quad r = \log K, \quad c = \frac{\pi^2}{6\alpha^2}. \quad (2.1)$$

Theorem 2.1 (All-orders logarithmic expansion; sources 40 and 48). *There are explicit polynomials $P_j(c) \in \mathbb{Q}[c]$ such that, for every fixed integer $R \geq 1$,*

$$\log a_\alpha(n) = \alpha K \left(r - 1 + \sum_{j=1}^R \frac{P_j(c)}{r^j} + O_{\alpha,R}(r^{-R-1}) \right). \quad (2.2)$$

The first seven are

$$\begin{aligned} P_1(c) &= c, & P_2(c) &= c, \\ P_3(c) &= c - \frac{1}{2}c^2, & P_4(c) &= c + \frac{9}{10}c^2, \\ P_5(c) &= c + \frac{41}{5}c^2 + \frac{1}{2}c^3, \\ P_6(c) &= c + 27c^2 + \frac{1973}{210}c^3, \\ P_7(c) &= c + \frac{129}{2}c^2 + \frac{8109}{70}c^3 - \frac{5}{8}c^4. \end{aligned} \quad (2.3)$$

Equations (7.9)–(8.3) below generate every P_j .

In particular,

$$\begin{aligned} \log a_\alpha(n) &= \alpha K(r - 1) + \frac{\pi^2 K}{6\alpha r} + \frac{\pi^2 K}{6\alpha r^2} \\ &\quad + \left(\frac{\pi^2}{6\alpha} - \frac{\pi^4}{72\alpha^3} \right) \frac{K}{r^3} + O_\alpha(K/r^4). \end{aligned} \quad (2.4)$$

This proves (1.3) for all five OEIS sequences, and more strongly identifies the $-\alpha K$ term with an error $O_\alpha(K/r)$.

Source 48 proves Theorem 2.1 independently (its Theorem 1.1, with $P_1, \dots, P_6$ written $c_1(A), \dots, c_6(A)$), and adds three remarks. First, its constants are uniform when α ranges over a fixed compact subinterval of $(0, \infty)$; they are *not* claimed to be uniform as $\alpha \downarrow 0$. Second, the case $R = 0$ states $\log a_\alpha(n) = \alpha K(r - 1) + O_\alpha(K/r)$, which already proves more than the displayed OEIS conjecture, and the first nonzero correction is positive:

$$\lim_{n \rightarrow \infty} \frac{r}{K} (\log a_\alpha(n) - \alpha K(r - 1)) = \frac{\pi^2}{6\alpha}. \quad (2.5)$$

Third, all-order statements in this report mean: *for each fixed truncation order, the asserted remainder bound holds*. No convergence of the infinite inverse-logarithmic series is assumed (source 48).

For a different level of precision, define

$$F(t) = \sum_{k \geq 1} \log(1 + k^\alpha e^{-tk}), \quad t > 0, \quad (2.6)$$

and let $t = t_n$ be the unique solution of $-F'(t) = n$. Set

$$s = -W_{-1}(-t/\alpha), \quad M = e^s, \quad t = \alpha s/M, \quad H = \frac{M}{\alpha(s - 1)}. \quad (2.7)$$

These definitions use the large real branch $s > 1$, valid for all sufficiently large n . Let $B = F''(t)$ and $\kappa_j = (-1)^j F^{(j)}(t)$.

Theorem 2.2 (Exact-saddle coefficient expansion; sources 40 and 48). *As $n \rightarrow \infty$,*

$$M \sim K, \quad s \sim r, \quad B \sim \frac{M^3}{\alpha s}, \quad a_\alpha(n) \sim \frac{\exp(F(t_n) + nt_n)}{\sqrt{2\pi B}}.$$

More precisely, for every fixed $J \geq 1$,

$$a_\alpha(n) = \frac{\exp(F(t_n) + nt_n)}{\sqrt{2\pi B}} \left(1 + \sum_{\ell=1}^{J-1} \mathcal{E}_\ell + O_{\alpha,J}((s/M)^J) \right), \quad (2.8)$$

where each $\mathcal{E}_\ell = O_{\alpha,\ell}((s/M)^\ell)$ is given explicitly in (6.2). The first correction is

$$\mathcal{E}_1 = \frac{\kappa_4}{8B^2} - \frac{5\kappa_3^2}{24B^3}. \quad (2.9)$$

Theorem 2.3 (Asymptotic inversion; sources 40 and 48). *For $Y \rightarrow \infty$, let $N_\alpha(Y)$ be the least integer n such that $a_\alpha(n) \geq e^Y$. Define*

$$w = W_0\left(\frac{Y}{\alpha e}\right), \quad R_0 = 1 + w, \quad K_0 = \frac{Y}{\alpha w}. \quad (2.10)$$

Then

$$N_\alpha(Y) = \frac{K_0^2}{2} \left(1 - \frac{2c}{R_0^2} - \frac{2c}{R_0^3} + \frac{4c^2 - 2c}{R_0^4} + O_\alpha(R_0^{-5}) \right). \quad (2.11)$$

A formal recurrence in Section 9 gives all inverse-logarithmic orders, with a remainder at every fixed order.

Theorem 2.4 (Logistic boundary and Gumbel extreme; sources 40 and 48). *Give $\mathcal{D}_n$ the probability law $\mathbb{P}_n(\lambda) = N(\lambda)^\alpha / a_\alpha(n)$. For each fixed $x \in \mathbb{R}$, a part near $M + xH$ has limiting occupation probability $(1 + e^x)^{-1}$; any fixed finite collection of distinct such sites is asymptotically independent. If L_n is the largest part, define*

$$b = b(t_n) > \alpha/t_n, \quad \frac{b^\alpha e^{-t_n b}}{t_n} = 1. \quad (2.12)$$

Then, for every fixed real x ,

$$\mathbb{P}_n\{t_n(L_n - b) \leq x\} \longrightarrow \exp(-e^{-x}), \quad \frac{L_n}{\sqrt{2n}} \longrightarrow 1 + \frac{1}{\alpha} \quad \text{in probability.} \quad (2.13)$$

A precision warning. The error in (2.2) is $O(K/r^{R+1})$, which still grows without bound for fixed R . Consequently, exponentiating a fixed truncation need not approximate $a_\alpha(n)$ to relative error $o(1)$. Formula (2.8) does, because it retains the *exact* saddle and the *exact* infinite sums. There is also a smaller exponential scale from noncentral Fourier arcs; Section 11 identifies it. We do not conflate these three statements into an unjustified single explicit coefficient transseries.

Source 48 draws the same line: the inverse-logarithmic expansion determines the complete block on the scales $Kr, K, K/r, K/r^2, \dots$; it does *not* resolve every smaller block or every exponentially small contribution, and a finite truncation of this block generally leaves a diverging error in $\log a_\alpha(n)$. A multiplicative equivalent is instead obtained from the exact saddle and exact convergent sums. Its numerical routines are independent checks, not substitutes for the proofs; they use arbitrary-precision arithmetic but not directed-rounding interval certification (source 48).

3 A short proof of all five posted conjectures

This section already proves the formulas that are explicitly conjectural in the OEIS entries. Later sections supply the stronger refinements.

Lemma 3.1 (A useful continuum estimate). *For $s \rightarrow \infty$, $M = e^s$, and $t = \alpha s/M$,*

$$F(t) = \alpha M(s/2 - 1) + O_\alpha(M/s). \quad (3.1)$$

Proof. The function $f_t(x) = \log(1 + x^\alpha e^{-tx})$ is nonnegative and unimodal on $(0, \infty)$, tends to zero at both ends, and has maximum $O_\alpha(s)$. For a nonnegative unimodal function, comparison on unit intervals gives

$$\left| \sum_{k \geq 1} f_t(k) - \int_0^\infty f_t(x) dx \right| \leq 2 \sup_x f_t(x) = O_\alpha(s).$$

With $x = Mu$, the exponent is $\alpha[s(1-u) + \log u]$. Replacing $\log(1 + e^z)$ by $\max(z, 0)$ changes the integral by $O_\alpha(M/s)$: in a fixed neighborhood of $u = 1$, the derivative of $s(u-1) - \log u$ is comparable with s , and $\int_{\mathbb{R}} \log(1 + e^{-|v|}) dv < \infty$. Outside that neighborhood, split off $0 < u < e^{-s/2}$; its integral is exponentially small times a polynomial in s . On the remaining region away from 1, the absolute exponent grows linearly in s , except for the negligible small lower crossing. Thus the remaining error is smaller than M/s . The lower crossing changes the following signed baseline by $O(1)$:

$$\alpha M \int_0^1 (s(1-u) + \log u) du = \alpha M(s/2 - 1).$$

The lattice error $O(s)$ is also $O(M/s)$. □

Proposition 3.2 (sources 40 and 48). *For every fixed $\alpha > 0$,*

$$\log a_\alpha(n) = \alpha \sqrt{2n} (\log \sqrt{2n} - 1) + O_\alpha \left(\frac{\sqrt{n}}{\log n} \right). \quad (3.2)$$

Proof. Let m be maximal with $m(m+1)/2 \leq n$, and write $d = n - m(m+1)/2$, so $0 \leq d \leq m$. The distinct partition

$$1 + 2 + \dots + (m-1) + (m+d)$$

has norm at least $m!$. Therefore

$$\log a_\alpha(n) \geq \alpha \log(m!) = \alpha K(r-1) + O_\alpha(r),$$

where $m = K + O(1)$ and the usual integral bounds, or Stirling's formula, give $\log(m!) = m \log m - m + O(\log m)$.

For the upper bound, nonnegative coefficients give $a_\alpha(n) \leq \exp(F(t) + nt)$ for every $t > 0$. Choose $t = \alpha r/K$, that is, $M = K$ and $s = r$ in Lemma 3.1. Since $n = K^2/2$,

$$F(t) + nt = \alpha K(r/2 - 1) + \alpha K r/2 + O_\alpha(K/r) = \alpha K(r-1) + O_\alpha(K/r).$$

The two bounds prove (3.2). $\square$

Remark 3.3. The lower-bound partition need not maximize the norm. An exact maximizer is unnecessary. Also, a bare asymptotic equivalence to $\alpha K(r-1)$ would not by itself certify $-\alpha K$ as the second term; the error in (3.2) does certify it.

3.1 Two structural facts and a second proof (source 48)

Why an expansion at the origin cannot simply be used at the saddle.

Proposition 3.4 (source 48). *The Taylor series of the analytic logarithm of A_α at $q = 0$ has radius of convergence $3^{-\alpha/3}$, although A_α is analytic throughout $|q| < 1$.*

Proof. The zeros of $1 + k^\alpha q^k$ have modulus $k^{-\alpha/k}$. Among positive integers k , the maximum of $\log k/k$ occurs at $k = 3$: the function decreases for $k > e$, and $\log 3/3 > \log 2/2$. Hence the closest zeros of the product have modulus $3^{-\alpha/3}$. The remaining factors converge normally and cannot remove a zero. Inside this circle all factors are nonzero, so the product has a single-valued analytic logarithm with value zero at the origin. At a closest zero its logarithm cannot be analytic. This proves the assertion. $\square$

Consequently,

$$\log A_\alpha(q) = \sum_{k \geq 1} \sum_{j \geq 1} \frac{(-1)^{j+1}}{j} k^{\alpha j} q^{kj}$$

is useful as a *formal* identity, but this expansion is not convergent at the positive saddles approaching $q = 1$. Treating it as a convergent expansion there would lose the transition that controls this problem.

The largest individual weight. Let

$$\Pi_\alpha(n) = \max_{\lambda \in \mathcal{D}_n} N(\lambda)^\alpha.$$

Proposition 3.5 (source 48). *For fixed $\alpha > 0$, with $K = \sqrt{2n}$ and $r = \log K$,*

$$\log \Pi_\alpha(n) = \alpha K(r-1) + O_\alpha(r). \quad (3.3)$$

Proof. Write $n = m(m+1)/2 + d$ with $0 \leq d \leq m$. The distinct partition $(1, 2, \dots, m-1, m+d)$ has weight $((m-1)!(m+d))^\alpha$. Since $m = K + O(1)$, Stirling's estimate yields the required lower bound, with error $O_\alpha(r)$.

For the upper bound set $t = \alpha r/K$. Any distinct partition λ of n satisfies

$$\alpha \sum_{k \in \lambda} \log k \leq tn + \sum_{k \geq 1} (\alpha \log k - tk)_+, \quad u_+ = \max(u, 0).$$

For sufficiently large K , the positive terms are exactly those with $2 \leq k < K$, apart from a possible zero term at an integer endpoint. Summing $\log k$ by Stirling's formula and k by the arithmetic progression formula gives

$$\sum_{k \geq 1} (\alpha \log k - tk)_+ = \alpha K(r/2 - 1) + O_\alpha(r).$$

Adding $tn = \alpha Kr/2$ proves the upper bound. $\square$

Lemma 3.6 (transition width; source 48). *Let $M = e^s$, $t = \alpha s/M$, and $s \rightarrow \infty$. Then*

$$\sum_{k \geq 1} \exp(-|\alpha \log k - tk|) = O_\alpha(M/s). \quad (3.4)$$

Proof. Put $\psi(u) = \alpha \log u - tu$. On $k \geq M/2$, the derivative of ψ is at most $-t/2$ once $s \geq 4$, and $\psi(M) = 0$. Thus this part of the sum is bounded by a two-sided geometric sum of scale $1/t = O_\alpha(M/s)$.

For $1 \leq k \leq M^{1/4}$, use the bound 1, giving $O(M^{1/4})$. On $[M^{1/4}, M/2]$, concavity of ψ implies that its minimum is at an endpoint. For large s , both endpoint values exceed $c_\alpha s$ for a constant $c_\alpha > 0$. This range therefore contributes $O_\alpha(M^{1-c_\alpha})$. Both estimates are $o_\alpha(M/s)$. Adjusting integer endpoints changes nothing. $\square$

Second proof of Proposition 3.2 (source 48). The lower bound follows from $a_\alpha(n) \geq \Pi_\alpha(n)$ and Proposition 3.5 (the same partition as in the first proof). For the upper bound, positivity gives $a_\alpha(n) \leq e^{tn} A_\alpha(e^{-t})$. Use $t = \alpha r/K$ and the exact identity

$$\log(1 + e^u) = u_+ + \log(1 + e^{-|u|}).$$

The positive-part sum was evaluated in the proof of Proposition 3.5; the second sum is bounded by Lemma 3.6 with $M = K$, $s = r$. We obtain

$$tn + F(t) = \alpha K(r - 1) + O_\alpha(K/r + r),$$

and $r = o(K/r)$. This proves (3.2) without any local limit theorem, continuum integral or formal asymptotic manipulation. $\square$

This argument is a separate, elementary proof of the OEIS conjecture; the remaining sections determine what lies inside its error term (source 48).

4 The exact probabilistic saddle and its scales

For fixed $t > 0$, let X_k be independent Bernoulli random variables with

$$p_k = \mathbb{P}_t(X_k = 1) = \frac{k^\alpha e^{-tk}}{1 + k^\alpha e^{-tk}}, \quad S = \sum_{k \geq 1} k X_k. \quad (4.1)$$

The sum is finite almost surely because $\sum p_k < \infty$. Multiplying the Bernoulli probabilities gives the exact identity

$$\mathbb{P}_t(S = n) = a_\alpha(n) e^{-nt - F(t)}. \quad (4.2)$$

Differentiation under the absolutely convergent sums shows

$$\mathbb{E}_t S = -F'(t) = \sum k p_k, \quad \text{Var}_t S = F''(t) = \sum k^2 p_k (1 - p_k) > 0.$$

The mean is continuous and strictly decreasing from ∞ to 0 as t increases from 0 to ∞ . Hence the saddle equation has a unique solution for every $n > 0$.

The large crossing $k^\alpha e^{-tk} = 1$ occurs at $k = M$ in (2.7). Its logarithmic derivative there is

$$\frac{\alpha}{M} - t = -\frac{\alpha(s-1)}{M} = -H^{-1}.$$

Thus H is the width of the transition from predominantly occupied to predominantly unoccupied sites.

Lemma 4.1 (Transition estimates). *For fixed $\alpha > 0$, as $s \rightarrow \infty$,*

$$\sum_{k \geq 1} k p_k \sim M^2/2, \quad (4.3)$$

$$\sum_{k \geq 1} k^2 p_k (1 - p_k) \sim M^2 H \sim M^3/(\alpha s). \quad (4.4)$$

For each fixed integer $j \geq 2$,

$$\sum_{k \geq 1} k^j p_k (1 - p_k) = O_{\alpha,j}(M^{j+1}/s). \quad (4.5)$$

Moreover, for every fixed nonnegative integer h ,

$$\frac{1}{H^{h+1}} \sum_{k \geq 1} |k - M|^h p_k (1 - p_k) \longrightarrow \int_{\mathbb{R}} |x|^h \frac{e^x}{(1 + e^x)^2} dx. \quad (4.6)$$

Proof. For $k = M + xH$ with x in a fixed bounded interval,

$$\alpha \log k - tk = \alpha \log(1 + xH/M) - \alpha s x H/M = -x + O_{\alpha,x}(s^{-2}).$$

Here and below rounding k to an integer changes the scaled coordinate by $O(H^{-1}) = o(1)$. Thus p_k converges to $(1 + e^x)^{-1}$. Since $H \rightarrow \infty$, sums over any fixed x -interval are Riemann sums. The limiting variance density has integral 1.

We record a tail justification so that this argument is not just a local heuristic. On $k/M \in [1/2, 3/2]$, the derivative of the log odds is negative and comparable to $-1/H$ for large s ; the variance density is therefore bounded by $C_\alpha e^{-c_\alpha |k-M|/H}$. On $k \geq 3M/2$, the odds themselves are exponentially small in s and then in tk . On $k \leq M/2$, use $p_k(1 - p_k) \leq \min(1, k^{-\alpha} e^{tk})$, and split at M/s . The range $M/s < k \leq M/2$ is exponentially small in s after scaling by any of the powers of M above. On $k \leq M/s$, the crude bound is $O_\alpha(k^{-\alpha})$; its contribution to (4.6), even with $|k - M|^h \leq M^h$, is $o(H^{h+1})$. Indeed the resulting ratio is bounded by a fixed power of s times $M^{-\min(\alpha, 1)}$, with an extra logarithm when $\alpha = 1$. These bounds establish uniform integrability for every fixed moment. They also prove (4.5) and (4.4).

Finally, $p_{\lfloor Mu \rfloor} \rightarrow 1$ for $0 < u < 1$ and tends to zero for $u > 1$. The same tail bounds and a Riemann sum give $M^{-2} \sum k p_k \rightarrow \int_0^1 u du = 1/2$. $\square$

Source 48 sets up the same Bernoulli representation (its (4.2) is identical, with the total written X_t and the variance V), and the same transition coordinate $s = -W_{-1}(-t/\alpha)$, which it calls the *upper transition coordinate*; the lower real branch is necessary because $s \rightarrow \infty$, and the branch conventions agree with the DLMP [15]. Its version of Lemma 4.1 (its Lemma 3.1) states $-F'(t) \sim M^2/2$, $B \sim M^3/(\alpha s)$ and $|\kappa_j| = O_{\alpha,j}(M^{j+1}/s)$ for $j \geq 2$, with the following proof.

Second proof of (4.3)–(4.5) (source 48). In the scaled variable $k = Mu$, the log odds equal $\alpha s(1 - u) + \alpha \log u$. For $0 < u < 1$ this tends to $+\infty$, and for $u > 1$ it tends to $-\infty$. Splitting the sum at fixed fractions of M , and using exponential decay above a fixed multiple of M , gives $M^{-2} \sum k p_k \rightarrow \int_0^1 u \, du$.

Near $k = M$, put $v = t(k - M)$. The log odds are

$$-v + \alpha \log \left(1 + \frac{v}{\alpha s} \right) \rightarrow -v.$$

The mesh in v is $t \rightarrow 0$. The derivative estimate used in Lemma 3.6 supplies an exponentially integrable bound on $k \geq M/2$. Thus a Riemann-sum limit gives

$$B \sim \frac{M^2}{t} \int_{-\infty}^{\infty} \frac{e^v}{(1 + e^v)^2} \, dv = \frac{M^2}{t} = \frac{M^3}{\alpha s}.$$

The range below $M/2$ is negligible. To see this for any fixed moment order, split it at $M^{1/4}$ as in Lemma 3.6, now multiplying by k^j . The small range is $O(M^{(j+1)/4})$; the other range is $O_\alpha(M^{j+1-c_\alpha})$. Above $M/2$, the exponential envelope gives $\sum k^j p_k(1 - p_k) = O_{\alpha,j}(M^{j+1}/s)$. This proves the cumulant bound, and normalization by the variance proves (4.8). $\square$

At the exact saddle, (4.3) gives $M \sim K$ and $s \sim r$. For cumulants, write $b_1(p) = p$ and $b_{j+1}(p) = p(1 - p)b'_j(p)$. Then

$$\kappa_j = \sum_{k \geq 1} k^j b_j(p_k), \quad |b_j(p)| \leq C_j p(1 - p) \quad (j \geq 2). \quad (4.7)$$

Consequently, with $\varepsilon = s/M$,

$$\frac{|\kappa_j|}{B^{j/2}} = O_{\alpha,j}(\varepsilon^{(j-2)/2}), \quad j \geq 3. \quad (4.8)$$

These bounds are deliberately conservative; signed cancellations can make individual corrections smaller. They suffice for all fixed Edgeworth orders.

5 The local limit theorem, including noncentral arcs

5.1 An elementary consecutive-block inequality

Lemma 5.1 (sources 40 and 48). *There is an absolute constant $c_0 > 0$ such that for every integer $h \geq 2$, every integer a , and every $|\theta| \leq \pi$,*

$$\sum_{k=a}^{a+h-1} \sin^2(k\theta/2) \geq c_0 \min(h, h^3\theta^2). \quad (5.1)$$

Proof. The geometric-sum formula gives

$$\sum_{k=a}^{a+h-1} \sin^2(k\theta/2) \geq \frac{1}{2} \left(h - \left| \frac{\sin(h\theta/2)}{\sin(\theta/2)} \right| \right).$$

When $h|\theta| \leq 1$, Taylor bounds for sine show that the difference in parentheses is at least $ch(h^2 - 1)\theta^2$. For $h|\theta| \geq 2\pi$, the bound $|\sin(\theta/2)| \geq |\theta|/\pi$ makes the ratio at most $h/2$. On the intermediate range $1 \leq h|\theta| \leq 2\pi$, its ratio to h is uniformly less than one. To see the uniformity, let $h \rightarrow \infty$ and use uniform convergence to $|\sin(x/2)/(x/2)| < 1$ on $1 \leq x \leq 2\pi$; the finitely many remaining h have the same strict inequality by the strict triangle inequality for a nonconstant geometric progression. Taking the minimum of the resulting positive constants proves the lemma. $\square$

Source 48 states the lemma for every sufficiently large h in the equivalent form $\sum_{k=a}^{a+h-1} \sin^2(k\theta/2) \geq c_0 h \min\{1, h^2\theta^2\}$.

Second proof (source 48). The sum is at least

$$\frac{1}{2} \left(h - \left| \sum_{k=a}^{a+h-1} e^{ik\theta} \right| \right) = \frac{1}{2} \left(h - \frac{|\sin(h\theta/2)|}{|\sin(\theta/2)|} \right).$$

For $h|\theta| \leq 1$, Taylor's formula bounds the deficit below by a constant times $h^3\theta^2$. For $1 \leq h|\theta| \leq C$, divide by h and use the strict inequality $|\sin u| < |u|$ away from zero, uniformly on a compact interval; the denominator converges uniformly to $|h\theta|/2$ in this range. For $h|\theta| \geq C$, use $|\sin(\theta/2)| \geq |\theta|/\pi$, choosing a fixed sufficiently large C . These three ranges prove the result. $\square$

5.2 Fourier decay and the central Gaussian

Let $\phi_t(\theta) = \mathbb{E}_t e^{i\theta(S - \mathbb{E}_t S)}$. The modulus identity for one Bernoulli factor implies

$$|\phi_t(\theta)| \leq \exp \left(-2 \sum_{k \geq 1} p_k(1-p_k) \sin^2(k\theta/2) \right). \quad (5.2)$$

In a consecutive block of length $h \asymp_\alpha M/s$ centered at M , all $p_k(1-p_k)$ are bounded below by a positive constant. On $|\theta| \leq c_1/M$, this block gives

$$|\phi_t(\theta)| \leq e^{-c_2 B \theta^2}. \quad (5.3)$$

Indeed $k \asymp M$ on the block, and $\sin(k\theta/2)$ is comparable with $k\theta$ for sufficiently small c_1 . On its complement in $[-\pi, \pi]$, Lemma 5.1 gives

$$|\phi_t(\theta)| \leq \exp(-c_3 M/s^3), \quad c_1/M \leq |\theta| \leq \pi. \quad (5.4)$$

All constants here may depend on α . Notice the power s^3 , rather than s : the random part sizes lie in a relatively narrow consecutive block, so one must control possible near-resonances. An argument checking only a neighborhood of zero would leave a genuine gap. Source 48 proves the same two bounds from the same block (its (5.3) and (5.4) are written with V for B and L for s): the worst case of Lemma 5.1 at $|\theta| = c_1/M$ is a constant times $h^3/M^2 \asymp_\alpha M/s^3$, so in particular no other root-of-unity arc contributes at an algebraic order.

At the saddle, Fourier inversion gives

$$\mathbb{P}_t(S = n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \phi_t(\theta) d\theta.$$

For bounded $x = \theta\sqrt{B}$, Taylor expansion and (4.8) give $\phi_t(x/\sqrt{B}) \rightarrow e^{-x^2/2}$. The central bound dominates this integrand by e^{-cx^2} . The noncentral integral, even after multiplication by $\sqrt{B}$, is $O(\sqrt{B}e^{-c_3 M/s^3}) = o(1)$. Thus

$$\mathbb{P}_t(S = n) \sim (2\pi B)^{-1/2}. \quad (5.5)$$

Together with (4.2), this proves the leading part of Theorem 2.2. In particular,

$$\log a_\alpha(n) = F(t_n) + nt_n + O_\alpha(s), \quad (5.6)$$

since $\log B = O(s)$.

5.3 A uniform variant used for conditioning

The same proof gives the following useful extension. Delete either a fixed number of sites $k = O(M)$, or all sites above a cutoff m for which $m/M \rightarrow \gamma > 1$ and the deleted mean and variance are $O(M)$ and $O(M^2)$. The central consecutive block remains available, the retained variance is $B(1 + o(1))$, and the cumulant bounds remain valid. If an integer target differs from the retained mean by $o(\sqrt{B})$, its point probability is

$$(2\pi B)^{-1/2}(1 + o(1)). \quad (5.7)$$

This follows in the central integral by inserting $e^{-ix(m-\mathbb{E}S)/\sqrt{B}} \rightarrow 1$; all the dominating estimates are unchanged. In the fixed-site deletion case, the deleted variance is automatically $O(M^2)$. The assertion is uniform for boundedly many deletions and shifts bounded by a fixed multiple of M . Source 48 records the same extension from its proof of Theorem 2.2, with a factor $e^{-ih\theta}$ in the integral: uniformly for $h = o(\sqrt{B})$,

$$\mathbb{P}_t(S = -F'(t) + h) = \frac{1 + o(1)}{\sqrt{2\pi B}} \quad \text{whenever } -F'(t) + h \text{ is an integer,} \quad (5.8)$$

also after deleting finitely many summands, or after truncating well above the active interval, provided the removed variance is $o(B)$ (source 48).

6 The all-orders coefficient correction

The normalized Fourier integrand has the expansion

$$\phi_t(x/\sqrt{B}) = e^{-x^2/2} \exp \left(\sum_{j \geq 3} \frac{\kappa_j (ix)^j}{j! B^{j/2}} \right). \quad (6.1)$$

Assign weight $j - 2$ to the j th normalized cumulant. Every monomial of odd weight has odd total power of x and integrates to zero. A monomial of weight 2ℓ has order $O(\varepsilon^\ell)$ by (4.8). Gaussian integration therefore gives

$$\mathcal{E}_\ell = \sum_{\substack{m_3, \dots, m_{2\ell+2} \geq 0 \\ \sum_{j=3}^{2\ell+2} (j-2)m_j = 2\ell}} \frac{(-1)^{T/2} (T-1)!!}{B^{T/2}} \prod_{j=3}^{2\ell+2} \frac{\kappa_j^{m_j}}{(j!)^{m_j} m_j!}, \quad T = \sum_{j=3}^{2\ell+2} j m_j. \quad (6.2)$$

Here T is automatically even. Besides (2.9), the next term is

$$\begin{aligned} \mathcal{E}_2 = & -\frac{\kappa_6}{48B^3} + \frac{7\kappa_3\kappa_5}{48B^4} + \frac{35\kappa_4^2}{384B^4} \\ & - \frac{35\kappa_3^2\kappa_4}{64B^5} + \frac{385\kappa_3^4}{1152B^6}. \end{aligned} \quad (6.3)$$

For completeness, the formal expansion above has the needed integrated remainder. On $|\theta| \leq c/M$, derivatives of order $j \geq 2$ of $\log \mathbb{E}_t e^{i\theta S}$ have absolute value at most $C_{\alpha,j} M^{j+1}/s$. For $k \leq 2M$, take c sufficiently small so that the Bernoulli denominators have modulus comparable with $1 + k^\alpha e^{-tk}$. For $k > 2M$, the odds tend uniformly to zero; the denominators are bounded away from zero and the exponentially decreasing tails give the same bound. These observations justify Taylor's formula with controlled remainders.

Fix J , choose a small $\delta > 0$ depending on J , and first restrict the normalized integral to $|x| \leq \varepsilon^{-\delta}$. Expand through weight $2J - 1$, using Taylor's formula for the exponential. The

terms of weight $2J$ and higher have an integrable bound of the form $C_J \varepsilon^J (1 + |x|^{C_J}) e^{-x^2/4}$, after choosing δ small enough. The omitted odd-weight terms integrate to zero on the symmetric interval. Between this interval and $c\sqrt{B}/M$, (5.3) makes the integral smaller than every power of ε . The remaining Fourier range is bounded by (5.4), with the same conclusion. This proves (2.8) and its error for every fixed J .

Remark 6.1 (What an effective finite-size bound would contain). Before absorbing flat terms into the asymptotic O -term, the proof contains an additional relative bound of the shape $C_\alpha \sqrt{B} \exp(-c_\alpha M/s^3)$ for noncentral arcs. This can be much larger than the central Edgeworth corrections at attainable n , especially for larger α . The theorem is not a claim that its first few terms have small relative error uniformly in α or for all moderate n .

6.1 The cumulative tuple form and a second proof (source 48)

Source 48 writes the corrections cumulatively. For a finitely supported tuple $\mathbf{m} = (m_3, m_4, \dots)$ of nonnegative integers put $w(\mathbf{m}) = \sum_{j \geq 3} (j-2)m_j$ and $T(\mathbf{m}) = \sum_{j \geq 3} j m_j$. Then, for every $J \geq 1$,

$$1 + \sum_{\ell=1}^{J-1} \mathcal{E}_\ell = \sum_{\substack{w(\mathbf{m}) < 2J \\ T(\mathbf{m}) \text{ even}}} (-1)^{T(\mathbf{m})/2} (T(\mathbf{m}) - 1)!! \prod_{j \geq 3} \frac{1}{m_j!} \left(\frac{\kappa_j}{j! B^{j/2}} \right)^{m_j}, \quad (6.4)$$

where the empty tuple contributes 1 and $(-1)!! = 1$. This is (6.2) summed over $\ell < J$ (an even T is the same as an even weight), and (2.8) reads $a_\alpha(n) = \frac{\exp(F(t_n) + n t_n)}{\sqrt{2\pi B}} ((6.4) + O_{\alpha,J}((s/M)^J))$; the leading formula is a genuine multiplicative equivalent.

Second proof of Theorem 2.2 (source 48). Fourier inversion and (4.2) reduce the assertion to the probability $\mathbb{P}_t(S = n)$ at the exact saddle. Set $u = \theta\sqrt{B}$ and center the characteristic function; its expansion at zero is (6.1), interpreted to any fixed finite Taylor order with a remainder.

The Taylor expansion is uniform for the following reasons. For $|\theta| \leq c_0/M$, derivatives of order $j \geq 2$ of the logarithm of the characteristic function are $O_{\alpha,j}(M^{j+1}/s)$. For k in a fixed multiple of $[1, M]$, either the Bernoulli parameter is away from $1/2$, or the small phase keeps the factor uniformly away from zero. For k above that multiple of M , p_k is small and its factor is again uniformly away from zero. The derivatives in both ranges are bounded by $C_j k^j p_k(1 - p_k)$; summation uses Lemma 4.1. Taylor's integral remainder therefore obeys the corresponding bound from (4.8).

Explicit finite-order bookkeeping for the central integral: fix J , put $D = 2J + 2$, and write

$$Z(u) = \sum_{j=3}^D \frac{\kappa_j (iu)^j}{j! B^{j/2}}.$$

Taylor's remainder in the exponent is bounded by $C_{\alpha,J} \varepsilon^{(D-1)/2} |u|^{D+1}$. Choose $0 < \sigma < 1/6$ sufficiently small that also $\sigma < (D-1)/(2(D+1))$. On $|u| \leq \varepsilon^{-\sigma}$, both $Z(u)$ and this remainder tend uniformly to zero. Consequently replacing the exact exponential by $e^{-u^2/2} e^{Z(u)}$ has integrated error $O_{\alpha,J}(\varepsilon^{J+1/2})$, by the bound $|e^{a+b} - e^a| \leq |b| e^{|a|+|b|}$ and a fixed Gaussian moment.

Next expand e^Z through degree $2J$. Its remainder, integrated against the Gaussian, is again $O_{\alpha,J}(\varepsilon^{J+1/2})$: every coefficient in Z is $O_{\alpha,J}(\varepsilon^{1/2})$, so $|Z|^{2J+1}$ is bounded by that power of ε times a fixed polynomial in $|u|$. In the finite expansion of the retained powers, any monomial of weight at least $2J$ has integrated absolute size $O_{\alpha,J}(\varepsilon^J)$. Every monomial of smaller weight occurs in these powers, because each factor has positive weight and the largest required index

is at most $2J + 1$. Odd total degree integrates to zero on the symmetric central interval. The surviving terms are precisely those in (6.4). This argument uses only finite Taylor expansions and their remainders, not uniform convergence of an infinite series.

The remainder of $|\theta| \leq c_0/M$ is smaller than every power of ε by (5.3); the rest of $[-\pi, \pi]$ is likewise negligible by (5.4). Integrating each retained polynomial over the whole real line introduces only another negligible Gaussian tail. The Gaussian moment is zero for odd T and equals $(T - 1)!!$ for even T ; the phase i^T supplies the sign. At weight two the only possibilities are $m_4 = 1$ and $m_3 = 2$, giving exactly (2.9). $\square$

7 Continuum smoothing to every logarithmic order

7.1 The inverse at the occupation boundary

Define

$$h_s(u) = s(u - 1) - \log u. \quad (7.1)$$

Near $u = 1$ it is strictly increasing. Let $u_s(v)$ be its inverse with $u_s(0) = 1$, and put

$$U_m(s) = u_s^{(m)}(0). \quad (7.2)$$

The derivative operator along the inverse is

$$\frac{d}{dv} = \frac{1}{s - 1/u} \frac{d}{du}. \quad (7.3)$$

Starting with u and applying (7.3) repeatedly, then setting $u = 1$, gives an effective rational-function recursion. Equivalently, Lagrange inversion gives

$$U_m(s) = (m - 1)! [z^{m-1}] \left(\frac{z}{sz - \log(1 + z)} \right)^m. \quad (7.4)$$

The first required odd derivatives are

$$\begin{aligned} U_1(s) &= \frac{1}{s - 1}, \\ U_3(s) &= \frac{2s + 1}{(s - 1)^5}, \\ U_5(s) &= \frac{24s^3 + 58s^2 + 22s + 1}{(s - 1)^9}, \\ U_7(s) &= \frac{720s^5 + 3708s^4 + 4400s^3 + 1452s^2 + 114s + 1}{(s - 1)^{13}}. \end{aligned} \quad (7.5)$$

Write $\eta(z) = (1 - 2^{1-z})\zeta(z)$. The elementary even kernel $g(v) = \log(1 + e^{-|v|})$ has moments

$$\int_{\mathbb{R}} v^{2j} g(v) dv = 2(2j)! \eta(2j + 2), \quad j \geq 0. \quad (7.6)$$

Indeed expand the logarithm on $(0, \infty)$ and integrate each exponentially decaying term. Odd moments vanish.

Lemma 7.1 (Boundary smoothing expansion). *For every fixed $J \geq 1$,*

$$\int_0^\infty \log(1 + x^\alpha e^{-tx}) dx = M \left(\alpha(s/2 - 1) + \sum_{j=1}^J \frac{2\eta(2j)}{\alpha^{2j-1}} U_{2j-1}(s) + O_{\alpha,J}(s^{-2J-1}) \right), \quad (7.7)$$

where $t = \alpha s e^{-s}$. The expansion may be differentiated with respect to s any fixed number of times, retaining a remainder smaller than any prescribed inverse power of s by taking sufficiently many terms first.

Proof. After $x = Mu$, the integrand is $\log(1 + e^{-\alpha h_s(u)})$. The baseline is $\alpha M \int_0^1 [s(1-u) + \log u] du = \alpha M(s/2 - 1)$. The discrepancy between this signed baseline and the positive-part baseline is confined to the small lower crossing and is exponentially small in s before multiplication by M .

Choose a fixed small $\rho > 0$ and restrict initially to $|u - 1| < \rho$. Replacing the original integrand by its positive-part baseline leaves $g(\alpha h_s(u))$. On a symmetric v -interval $|v| < \gamma s$ inside the image of this neighborhood, changing variables $v = h_s(u)$ gives

$$\int g(\alpha v) u'_s(v) dv.$$

The omitted portions have exponentially small tails. Taylor-expand $u'_s(v)$ through degree $2J - 1$. On the chosen interval, repeated use of (7.3) bounds the derivative of order $2J$ of u'_s by $O_J(s^{-2J-1})$. The Taylor remainder is thus bounded by $C_J s^{-2J-1} |v|^{2J}$, whose product with $g(\alpha v)$ is integrable. After extending the polynomial terms to the whole real line, odd terms vanish, and (7.6) supplies exactly $2\eta(2j)U_{2j-1}(s)/\alpha^{2j-1}$.

Here is a more explicit treatment of the region near zero, which must not be mistaken for part of the main boundary. On $0 < u < e^{-s/2}$, both the original integrand and the absolute baseline have integral at most $C_\alpha s e^{-s/2}$. On $e^{-s/2} \leq u \leq 1 - \rho$, the concavity of $s(1-u) + \log u$ puts it above a positive multiple of s for large s . The smoothing error there is exponentially small. On $u \geq 1 + \rho$ the log odds are negative and decrease at rate comparable with s . All these contributions are $O_{\alpha,J}(s^{-2J-1})$ for every fixed J .

Differentiation with respect to s inserts powers of $1 - u$ and derivatives of the same logistic kernel. On the boundary interval the inverse derivatives still have polynomial bounds in s^{-1} ; away from it, exponential tail bounds remain after multiplication by any fixed polynomial. Taylor expansion to a higher order therefore gives the differentiated version with any prescribed inverse-power precision. No assumption about differentiating an arbitrary O -term is being used. $\square$

7.2 From the continuum to the lattice

As in Lemma 3.1, the difference between $F(t)$ and its continuum integral is $O_\alpha(s)$. We also need the mean. The function

$$x \mapsto \frac{x^{\alpha+1} e^{-tx}}{1 + x^\alpha e^{-tx}} = xp(x)$$

is unimodal and has maximum $O_\alpha(M)$. To verify unimodality, its derivative has the sign of $1 + (\alpha - tx)(1 - p(x))$. This is positive for $x \leq \alpha/t$. For $x > \alpha/t$, the function $(tx - \alpha)(1 - p(x))$ increases strictly from zero to infinity. Thus

$$\sum_{k \geq 1} k p_k = \int_0^\infty xp(x) dx + O_\alpha(M). \quad (7.8)$$

This exact lattice error is negligible at every fixed inverse-logarithmic order relative to the mean M^2 .

Introduce the formal Laurent series at $s = \infty$

$$d_\alpha(s) = \sum_{j \geq 1} \frac{2\eta(2j)}{\alpha^{2j}} U_{2j-1}(s). \quad (7.9)$$

The notation is asymptotic/formal: no convergence of the infinite sum in j is asserted. Each coefficient of s^{-1} uses only finitely many summands, since $U_m(s) = O_m(s^{-m})$. The expansion begins

$$d_\alpha(s) = \frac{c}{s} + \frac{c}{s^2} + \frac{c}{s^3} + \frac{c + 7c^2/5}{s^4} + \frac{c + 77c^2/10}{s^5} + \frac{c + 49c^2/2 + 372c^3/35}{s^6} + \dots \quad (7.10)$$

In the all-orders logarithmic sense,

$$F(t) \sim \alpha M(s/2 - 1 + d_\alpha(s)). \quad (7.11)$$

The lattice error $O(s)$ and the non-boundary continuum pieces are smaller than M/s^R for every fixed R . They are *not* thereby zero and must not be discarded when a multiplicative coefficient equivalent is wanted.

7.3 A second route: an explicit differential operator (source 48)

Source 48 proves the two lattice estimates by the bounded-variation inequality $|\sum_{k \geq 1} f(k) - \int_0^\infty f| \leq \int_0^\infty |f'| = 2 \max f$ for a nonnegative unimodal f with $f(0) = 0$ tending to zero at infinity, applied to $\log(1 + \xi^\alpha e^{-t\xi})$ (maximum $O_\alpha(s)$) and to $\xi p(\xi)$ (maximum $O_\alpha(M)$, unimodal by the argument before (7.8)). It then replaces the inverse u_s and the Lagrange coefficients U_m by one explicit operator. Let

$$f(s) = \frac{s}{s-1}, \quad \mathcal{T} = \frac{s}{s-1} \frac{d}{ds}, \quad (7.12)$$

with η as in Section 7, and define the asymptotic series at $s = \infty$

$$\boxed{\mathcal{C}_\alpha(s) \sim 2 \sum_{j \geq 0} \frac{\eta(2j+2)}{\alpha^{2j+2}} \mathcal{T}^{2j} f(s), \quad \mathcal{B}_\alpha(s) = \mathcal{C}_\alpha(s) + \mathcal{T} \mathcal{C}_\alpha(s).} \quad (7.13)$$

The formula is locally finite after expansion in s^{-1} : for $j \geq 1$, $\mathcal{T}^{2j} f(s) = O(s^{-2j-1})$, so only finitely many summands affect any prescribed coefficient.

Lemma 7.2 (upper-transition expansion; source 48). *With $t = \alpha s e^{-s}$, in the sense of asymptotic series to every fixed finite order,*

$$F(t) \sim \frac{\alpha^2}{t} \left(\frac{s^2}{2} - s + \mathcal{C}_\alpha(s) \right), \quad (7.14)$$

$$-F'(t) \sim \frac{\alpha^2}{t^2} \left(\frac{s^2}{2} + \mathcal{B}_\alpha(s) \right). \quad (7.15)$$

More precisely, replacing $\mathcal{C}_\alpha$ or $\mathcal{B}_\alpha$ by its polynomial through s^{-R} leaves $O_{\alpha,R}(s^{-R-1})$ inside the corresponding parentheses, for every fixed $R \geq 0$.

Since $\alpha^2/t = \alpha M/s$, (7.14) is (7.11) with $d_\alpha(s) = \mathcal{C}_\alpha(s)/s$, and (7.15) is the mean relation (8.1) with $Q_\alpha(s) = 1 + 2\mathcal{B}_\alpha(s)/s^2$.

Proof. In the continuum integral of F , with integration variable ξ , put $y = t\xi/\alpha$. Since $\log(\alpha/t) = s - \log s$, the exponent becomes

$$\alpha \log \xi - t\xi = \alpha \tilde{h}_s(y), \quad \tilde{h}_s(y) = s - y + \log(y/s).$$

The zero-temperature contribution is

$$\int_0^s \alpha \tilde{h}_s(y) dy = \alpha(s^2/2 - s). \quad (7.16)$$

Subtracting this reference integral is legitimate even though $\tilde{h}_s$ becomes negative very near zero; that region is included in the error estimate.

On $y \geq s/2$, use the increasing coordinate $v = y - s - \log(y/s) = -\tilde{h}_s(y)$. If $y = y(v)$ is its inverse, then $y(0) = s$ and $y'(v) = y(v)/(y(v) - 1) = f(y(v))$. The difference from the reference integrand is $g(\alpha v) = \log(1 + e^{-\alpha|v|})$, an even exponentially decreasing function, with moments

$$\int_{-\infty}^{\infty} v^{2j} g(\alpha v) dv = \frac{2(2j)!}{\alpha^{2j+1}} \eta(2j+2), \quad (7.17)$$

by (7.6); these are the elementary Fermi–Dirac moments underlying Sommerfeld expansions [16]. Since $d/dv = f(y) d/dy$, Taylor expansion of $y'(v)$ at zero gives

$$\left. \frac{d^{2j}}{dv^{2j}} y'(v) \right|_{v=0} = \mathcal{T}^{2j} f(s).$$

On $|v| \leq s/4$, $y(v) \asymp s$, and for each fixed $a \geq 1$, $\mathcal{T}^a f(y) = O_a(s^{-a-1})$. Expanding through the odd degree $2J+1$, integrating against the even kernel, and bounding the next derivative gives an error $O_{\alpha,J}(s^{-2J-3})$. The portions with $|v| > s/4$ are exponentially small, since $y'(v)$ stays bounded on the domain $y \geq s/2$.

The range $0 < y < s/2$ also has exponentially small error *inside the scaled integral*. Split it at $e^{-s/2}$. On the small interval, integrate the bound $C_\alpha(1 + s + |\log y|)$, obtaining $O_\alpha(se^{-s/2})$. On $[e^{-s/2}, s/2]$, concavity gives $\tilde{h}_s(y) \geq s/2 - \log s - O(1)$, so the discrepancy is $O_\alpha(s^{\alpha+1}e^{-\alpha s/2})$. These errors need not remain exponentially small in the original, unscaled variable, but they are smaller than every inverse power of s inside the present parentheses. Multiplying the integral by α/t , the moments (7.17) give (7.14); the sum–integral error is absorbed at every fixed order.

The mean formula is proved directly, rather than by differentiating an unspecified big- O term. The scaled mean integral is $\frac{\alpha^2}{t^2} \int_0^\infty y (1 + e^{\alpha v(y)})^{-1} dy$, with reference integral $\int_0^s y dy = s^2/2$. Under the same coordinate change the difference kernel is the odd function

$$\mathcal{K}_\alpha(v) = \frac{1}{1 + e^{\alpha v}} - \mathbf{1}_{v < 0},$$

integrated against $y(v)y'(v)$. Its odd moments are

$$\int_{-\infty}^{\infty} v^{2j+1} \mathcal{K}_\alpha(v) dv = \frac{2(2j+1)!}{\alpha^{2j+2}} \eta(2j+2).$$

The elementary operator identity $\mathcal{T}(sf(s)) = f(s) + \mathcal{T}f(s)$ shows that the resulting correction is precisely $\mathcal{C}_\alpha + \mathcal{T}\mathcal{C}_\alpha$. The same cutoff and Taylor estimates apply, and (7.8) absorbs the discretization error. This proves (7.15) with controlled remainders, independently of any differentiation rule for errors. $\square$

In the parameter c , the beginning of (7.13) is

$$\mathcal{C}_\alpha(s) = c + \frac{c}{s} + \frac{c}{s^2} + \frac{c + \frac{7}{5}c^2}{s^3} + \frac{c + \frac{77}{10}c^2}{s^4} + \frac{c + \frac{49}{2}c^2 + \frac{372}{35}c^3}{s^5} + O_\alpha(s^{-6}), \quad (7.18)$$

$$\mathcal{B}_\alpha(s) = c + \frac{c}{s} + \frac{-2c + \frac{7}{5}c^2}{s^3} + \frac{-5c + \frac{7}{2}c^2}{s^4} + O_\alpha(s^{-5}). \quad (7.19)$$

In particular, the coefficient of s^{-2} in $\mathcal{B}_\alpha$ is zero, and $\mathcal{C}_\alpha(s)/s$ reproduces (7.10) term by term. [Merge note, batch 7301.] The intake recomputed (7.18), (7.19), (8.8) and $P_1, \dots, P_6$ from (7.13) and (8.10) in exact rational arithmetic, for three values of c ; all agree.

8 Eliminating the saddle and generating all coefficients

Since $dt/ds = -\alpha(s-1)/M$, differentiating the continuum expansion and using (7.8) gives the formal mean relation

$$n \sim \frac{M^2}{2} Q_\alpha(s), \quad Q_\alpha(s) = 1 + \frac{2(d_\alpha(s) + d'_\alpha(s))}{s-1}. \quad (8.1)$$

Here d'_α denotes formal differentiation. Thus

$$r = s + \frac{1}{2} \log Q_\alpha(s). \quad (8.2)$$

Because $Q_\alpha(s) = 1 + O(s^{-2})$, this has a unique formal inverse $s = r + O(r^{-2})$. It is computed recursively, for example by iterating $s \leftarrow r - \frac{1}{2} \log Q_\alpha(s)$ in the ring of formal series in r^{-1} . Every iteration determines additional coefficients.

Substitution in $F(t) + nt$ gives

$$\mathcal{H}_\alpha(r) = \frac{s(Q_\alpha(s) + 1)/2 - 1 + d_\alpha(s)}{\sqrt{Q_\alpha(s)}}, \quad r = s + \frac{1}{2} \log Q_\alpha(s). \quad (8.3)$$

By construction,

$$\mathcal{H}_\alpha(r) = r - 1 + \sum_{j \geq 1} P_j(c) r^{-j}. \quad (8.4)$$

Equations (7.3), (7.9), (8.1), and (8.3) are an all-orders algorithm involving only rational operations and expansions of logarithms and square roots. Since even zeta values are rational multiples of powers of π^2 , the coefficients are polynomials over $\mathbb{Q}$ in c .

Proof of Theorem 2.1. Fix the desired order R . Apply Lemma 7.1 and its mean version with more than enough terms that both remainders, after all operations, are $O_{\alpha,R}(r^{-R-1})$ in the normalized entropy. The exact mean relation has an additional error $O(M)$, negligible compared with $M^2 r^{-A}$ for every fixed A . Inverting (8.2) is stable because its derivative is $1 + O(s^{-3})$. Finite Taylor substitution is consequently legitimate with remainders, not only as a formal manipulation.

We obtain

$$F(t_n) + nt_n = \alpha K \left(r - 1 + \sum_{j=1}^R P_j(c) r^{-j} + O_{\alpha,R}(r^{-R-1}) \right).$$

Equation (5.6) contributes only $O(s)$, which is $o(Kr^{-R-1})$. This proves (2.2). Direct expansion gives (2.3); the supplied symbolic script reproduces those coefficients. $\square$

One further coefficient for an independent check. The same algorithm gives

$$P_8(c) = c + \frac{259}{2}c^2 + \frac{7271}{10}c^3 + \frac{157813}{840}c^4. \quad (8.5)$$

The continuum quadrature script checks the approach of the remainder after P_7 , multiplied by r^8 , to this polynomial. It does not replace the preceding proof or evaluate integer coefficients at those enormous sizes.

8.1 The same elimination in operator form (source 48)

At the exact saddle, Lemma 7.2 gives, to every finite inverse-logarithmic order,

$$K \sim M \left(1 + \frac{2\mathcal{B}_\alpha(s)}{s^2} \right)^{1/2}, \quad (8.6)$$

meaning the full finite-order expansion, not merely a leading ratio. Equivalently, its formal inverse is

$$r = s + \frac{1}{2} \log \left(1 + \frac{2\mathcal{B}_\alpha(s)}{s^2} \right), \quad (8.7)$$

which is (8.2). There is a unique solution with $s = r + O(r^{-2})$ as a formal series in r^{-1} . Indeed, put $z = r^{-1}$ and $\nu = s - r$. Then $s^{-1} = z/(1 + z\nu)$. The right-hand side of the fixed-point equation

$$\nu = -\frac{1}{2} \log \left(1 + \frac{2\mathcal{B}_\alpha(r + \nu)}{(r + \nu)^2} \right)$$

has order z^2 , and changing ν by $O(z^d)$ changes the right-hand side by $O(z^{d+3})$. Iteration starting with $\nu = 0$ therefore determines every coefficient after finitely many steps. The first terms are

$$s = r - \frac{c}{r^2} - \frac{c}{r^3} + \frac{c^2}{r^4} + \frac{2c - \frac{7}{5}c^2}{r^5} + O_\alpha(r^{-6}). \quad (8.8)$$

At the saddle, addition of (7.14) and nt gives

$$F(t) + nt \sim \alpha M \left(s - 1 + \frac{2\mathcal{C}_\alpha(s) + \mathcal{T}\mathcal{C}_\alpha(s)}{s} \right). \quad (8.9)$$

The logarithm of the Gaussian prefactor in Theorem 2.2 is $O_\alpha(s)$, and the logarithm of its relative correction is $o(1)$; both are smaller than every scale K/r^R . Thus the formal series determining Theorem 2.1 is

$$\boxed{\mathcal{H}_\alpha(r) = \frac{s - 1 + (2\mathcal{C}_\alpha(s) + \mathcal{T}\mathcal{C}_\alpha(s))/s}{\sqrt{1 + 2\mathcal{B}_\alpha(s)/s^2}}, \quad \mathcal{H}_\alpha(r) = r - 1 + \sum_{j \geq 1} P_j(c)r^{-j}}, \quad (8.10)$$

with s substituted from (8.7). This is (8.3): with $d_\alpha = \mathcal{C}_\alpha/s$ and $Q_\alpha = 1 + 2\mathcal{B}_\alpha/s^2$ the numerators agree, since $s(Q_\alpha + 1)/2 - 1 + d_\alpha = s - 1 + (\mathcal{B}_\alpha + \mathcal{C}_\alpha)/s$.

Why the formal rule proves actual asymptotics. Fix a desired final order R . Take sufficiently many terms, for example through order s^{-R-4} , in the two estimates of Lemma 7.2. Their remainders are genuine uniform estimates on large real s . The derivative of the finite approximation in (8.7) is $1 + O(s^{-3})$, bounded away from zero. The mean value theorem transports its error to an error of the same inverse-log order. Substitution into (8.9), whose normalized derivative is bounded at the relevant order, then gives the remainder in (2.2). This is a finite calculation with controlled remainders, not an assumption that the infinite formal series converges. Together with Theorem 2.2, this is source 48's proof of Theorem 2.1; substitution of (7.18)–(8.8) into (8.10) gives $P_1, \dots, P_6$.

No special symbolic integration routine is needed. In the variable $z = s^{-1}$, the operator is

$$\mathcal{T} = -\frac{z^2}{1-z} \frac{d}{dz}, \quad f = \frac{1}{1-z}. \quad (8.11)$$

The factors $2\eta(2j+2)/\alpha^{2j+2}$ are rational multiples of c^{j+1} , since even zeta values are rational multiples of even powers of π . Truncated polynomial arithmetic with (8.11), followed by the fixed-point substitution, generates the coefficients at any prescribed finite order. The program `code/48-a022629-coefficients.py` implements this construction and verifies the six displayed polynomials and six inverse coefficients exactly.

8.2 An entropy factor beyond the largest summand (source 48)

Combining Theorem 2.1 with Proposition 3.5 gives an additional conclusion.

Corollary 8.1 (source 48).

$$\log \frac{a_\alpha(n)}{\Pi_\alpha(n)} \sim \frac{\pi^2}{6\alpha} \frac{K}{r}. \quad (8.12)$$

Thus the total weighted enumeration exceeds its largest individual weight by a factor that tends to infinity, and whose logarithm has a precise leading constant.

For $\alpha = 1$, this explains why using only the most heavily weighted partition gets the first two large terms right but misses a multiplicatively enormous factor. The transition contains approximately M/s available sizes, and its accumulated smoothing contribution produces this factor.

9 Lambert- W inversion with an error theorem

The coefficients are strictly increasing for $n \geq 1$. Indeed, increasing the largest part of a distinct partition by one is an injection into distinct partitions of $n + 1$, and strictly increases each positive norm weight. Thus the threshold index $N_\alpha(Y)$ in Theorem 2.3 is well-defined for large Y .

The dominant inversion problem is

$$Y = \alpha K(\log K - 1).$$

Writing $w = \log K - 1$ gives $we^w = Y/(\alpha e)$, proving (2.10). In particular $\log K_0 = R_0$. Set $K = K_0 E$. The all-orders formal inversion is the single equation

$$E \mathcal{H}_\alpha(R_0 + \log E) = R_0 - 1, \quad E = 1 + O(R_0^{-2}). \quad (9.1)$$

The derivative with respect to E is $R_0 + O(1)$ at $E = 1$, so successive coefficients are uniquely determined. In terms of c , they are

$$\begin{aligned} E = 1 - \frac{c}{R_0^2} - \frac{c}{R_0^3} + \frac{3c^2/2 - c}{R_0^4} - \frac{c + 2c^2/5}{R_0^5} \\ - \frac{c + 46c^2/5 + 5c^3/2}{R_0^6} + \dots, \end{aligned} \quad (9.2)$$

and hence

$$\begin{aligned} E^2 = 1 - \frac{2c}{R_0^2} - \frac{2c}{R_0^3} + \frac{4c^2 - 2c}{R_0^4} + \frac{6c^2/5 - 2c}{R_0^5} \\ - \frac{2c + 77c^2/5 + 8c^3}{R_0^6} + \dots. \end{aligned} \quad (9.3)$$

The asymptotic index is $K_0^2 E^2/2$.

To justify inversion of the sequence rather than just of a formal expression, fix an order, use Theorem 2.1 two orders farther than needed, and bracket the target by the corresponding smooth functions with the two signs of its error bound. Their derivatives with respect to K are $\alpha r(1 + o(1)) > 0$. An entropy error $O(Kr^{-R-1})$ therefore changes K by $O(Kr^{-R-2})$, and changes $n = K^2/2$ by $O(K^2 r^{-R-2})$. These bounds transfer the formal inversion to the actual threshold by monotonicity. Rounding the brackets to integers costs $O(1)$, smaller than all of these fixed-order error scales. This proves Theorem 2.3 and its all-orders extension.

Remark 9.1 (Two different Lambert branches). The saddle phase uses W_{-1} because $s \rightarrow \infty$ solves $se^{-s} = t/\alpha \rightarrow 0^+$. The threshold inverse uses W_0 at a positive argument. Interchanging the branches would destroy the relevant scale. If inversion in elementary logarithms is desired, the usual formal solution of $w + \log w = \log(Y/(\alpha e))$ can be substituted into (9.3); retaining W_0 is shorter and preserves more precision.

9.1 The same inversion in source 48's form (source 48)

Source 48 proves Theorem 2.3 independently (its Theorem 6.1), with the threshold written in the logarithm, $N_\alpha(Y) = \min\{n \geq 1 : \log a_\alpha(n) \geq Y\}$; for a threshold on the sequence itself one uses $N_\alpha(\log \cdot)$. The minimum is well defined because the coefficients diverge (Proposition 3.2) and increase strictly for $n \geq 1$ by the injection above; the initial equality $a_\alpha(0) = a_\alpha(1) = 1$ is immaterial. Its statement displays two more orders than (2.11):

$$N_\alpha(Y) = \frac{K_0^2}{2} \left(1 - \frac{2c}{R_0^2} - \frac{2c}{R_0^3} + \frac{4c^2 - 2c}{R_0^4} + \frac{\frac{6}{5}c^2 - 2c}{R_0^5} - \frac{8c^3 + \frac{77}{5}c^2 + 2c}{R_0^6} + O_\alpha(R_0^{-7}) \right), \quad (9.4)$$

the coefficients of (9.3); integer rounding changes the result by at most an additional $O(1)$, absorbed in every fixed inverse-logarithmic remainder. With $E = 1 + \delta$ and $C(r) = \sum_{j \geq 1} P_j(c)r^{-j}$, its normalized equation for the corrected inverse,

$$\delta = - \frac{(1 + \delta) \log(1 + \delta) - \delta + (1 + \delta)C(R_0 + \log(1 + \delta))}{R_0}, \quad (9.5)$$

is (9.1) solved for δ . Starting from $\delta = 0$, finite formal iteration gives the unique series with $\delta = O(R_0^{-2})$, whose coefficients are those of $E - 1$ in (9.2).

Second proof of the threshold transfer (source 48). The core equation $K(\log K - 1) = Y/\alpha$ is solved exactly by $K = K_0$, using $we^w = Y/(\alpha e)$. This is the principal, positive Lambert branch, distinct from the lower branch used for the saddle. Choose any finite truncation of Theorem 2.1 and define its continuous approximation on large real K . Its derivative is $\alpha(\log K + O_\alpha(1/\log K))$, positive and comparable to $\alpha \log K$. An error $O_{\alpha,R}(Kr^{-R-1})$ in the logarithm therefore moves its inverse by $O_{\alpha,R}(Kr^{-R-2})$. This follows directly from the mean value theorem on a fixed relative neighborhood of the core root; the preliminary leading asymptotic puts the true root in that neighborhood. For the discrete threshold, apply these upper and lower approximations at $N_\alpha(Y)$ and $N_\alpha(Y) - 1$. Adjacent values of $K = \sqrt{2n}$ differ by $O(1/K)$, so the final bracketing introduces only an $O(1)$ uncertainty in n . Taking the forward expansion to sufficiently high fixed order proves any desired inverse order. $\square$

The first inverse correction is negative: the actual enumeration is larger than its core approximation, so it reaches a specified threshold sooner. For A022629 ($\alpha = 1$, $2c = \pi^2/3$) this begins

$$N_1(Y) = \frac{1}{2} \left(\frac{Y}{W_0(Y/e)} \right)^2 \left(1 - \frac{\pi^2}{3(1 + W_0(Y/e))^2} + O((1 + W_0(Y/e))^{-3}) \right). \quad (9.6)$$

The distinction between inversion in $Y = \log a(n)$ and inversion in $a(n)$ is essential; replacing Y by the original threshold would give a wholly wrong scale (source 48).

10 Conditioned geometry: a logistic boundary and a Gumbel maximum

Conditioning the Bernoulli law (4.1) on $S = n$ gives exactly

$$\mathbb{P}_n(\lambda) = \frac{N(\lambda)^\alpha}{a_\alpha(n)}. \quad (10.1)$$

No approximation is involved in this identification.

10.1 The boundary profile

Fix distinct integer sites $k_{1,n}, \dots, k_{d,n}$ with $(k_{j,n} - M)/H \rightarrow x_j \in \mathbb{R}$. Their unconditioned probabilities tend to $p(x_j) = (1 + e^{x_j})^{-1}$ by Lemma 4.1. For any prescribed pattern $\epsilon_j \in \{0, 1\}$, independence gives

$$\begin{aligned} & \mathbb{P}_n(X_{k_{j,n}} = \epsilon_j \text{ for all } j) \\ &= \prod_{j=1}^d p_{k_{j,n}}^{\epsilon_j} (1 - p_{k_{j,n}})^{1-\epsilon_j} \frac{\mathbb{P}_t\{S^{\text{rem}} = n - \sum_j k_{j,n} \epsilon_j\}}{\mathbb{P}_t(S = n)}. \end{aligned} \quad (10.2)$$

The shifted target differs from the remaining mean by $O(M) = o(\sqrt{B})$. The uniform local limit theorem in Section 5.3 makes the ratio of point probabilities tend to one. This proves both the logistic limit and finite-site asymptotic independence. Similarly, a site at γM has limiting occupation 1 for $0 < \gamma < 1$, and 0 for $\gamma > 1$.

10.2 The correct centering of the maximum

The large solution of (2.12) exists uniquely for sufficiently small t . With $z = tb$, the equation is $z^\alpha e^{-z} = t^{\alpha+1}$, so

$$b(t) = -\frac{\alpha}{t} W_{-1} \left(-\frac{t^{1+1/\alpha}}{\alpha} \right). \quad (10.3)$$

Solving its logarithm, $z - \alpha \log z = -(\alpha + 1) \log t$, with $t = \alpha s e^{-s}$ yields

$$\frac{b}{M} = 1 + \frac{1}{\alpha} + \frac{-\log s + \alpha \log(\alpha + 1) - (\alpha + 1) \log \alpha}{\alpha s} + O_\alpha \left(\frac{\log s}{s^2} \right). \quad (10.4)$$

Keeping only $b \sim (1 + 1/\alpha)M$ is not adequate for the Gumbel centering: the omitted shift has size $M \log s/s$, larger than $1/t$. Keeping all the displayed terms in (10.4) does give an $o(1/t)$ centering error. The exact definition (2.12) or (10.3) avoids this truncation bookkeeping altogether.

Proof of the maximum assertion in Theorem 2.4. Fix $x \in \mathbb{R}$ and put $m = \lfloor b + x/t \rfloor$. Then $tm \rightarrow \infty$ and $m/M \rightarrow 1 + 1/\alpha > 1$. In the tail $k > m$, all odds tend uniformly to zero. A sum–integral comparison followed by integration by parts gives

$$\sum_{k>m} p_k \sim \sum_{k>m} k^\alpha e^{-tk} \sim \frac{m^\alpha e^{-tm}}{t} \rightarrow e^{-x}. \quad (10.5)$$

For the second equivalence, substitute $v = tx$ in the integral and use $\int_z^\infty v^\alpha e^{-v} dv = z^\alpha e^{-z}(1 + O_\alpha(z^{-1}))$. The endpoint discretization has relative size $O(t)$. The last limit follows from

$b^\alpha e^{-tb}/t = 1$ and $m/b \rightarrow 1$. Also, $\sup_{k>m} p_k = O(t)$, so $\sum_{k>m} p_k^2 = o(1)$. Therefore the unconditioned probability of no part above m is

$$\prod_{k>m} (1 - p_k) \longrightarrow e^{-e^{-x}}. \quad (10.6)$$

It remains to justify conditioning, rather than silently replacing one ensemble by another. The removed mean and variance satisfy

$$\sum_{k>m} kp_k = O(M), \quad \sum_{k>m} k^2 p_k (1 - p_k) = O(M^2).$$

These bounds follow from the same tail-integral calculation with one or two extra powers of k . The remaining sum $S_{\leq m}$ thus has mean $n - O(M)$ and variance $B(1 + o(1))$. Its central block around M is intact. The uniform local limit estimate gives

$$\frac{\mathbb{P}_t(S_{\leq m} = n)}{\mathbb{P}_t(S = n)} \longrightarrow 1.$$

Independence of the retained and removed sites now gives

$$\mathbb{P}_n(L_n \leq m) = \prod_{k>m} (1 - p_k) \frac{\mathbb{P}_t(S_{\leq m} = n)}{\mathbb{P}_t(S = n)} \longrightarrow e^{-e^{-x}}.$$

This proves the Gumbel limit. Its tightness, (10.4), $M \sim K$, and $(tM)^{-1} = 1/(\alpha s) \rightarrow 0$ imply $L_n/K \rightarrow 1 + 1/\alpha$ in probability. $\square$

The two locations are genuinely different. The occupation boundary lies near $M \sim \sqrt{2n}$, while the maximum lies near $(1 + 1/\alpha)M$. For $\alpha = 1$, a typical largest part is asymptotic to $2\sqrt{2n}$, not to $\sqrt{2n}$. Individual occupancies far above the boundary are small, but there are enough possible sites to move the extreme by a macroscopic multiple of M . This is why a boundary profile alone does not determine the largest part.

10.3 Explicit centring in $\log \sqrt{2n}$ (source 48)

Source 48 proves the Gumbel law independently, with the centring written in $K = \sqrt{2n}$ and $r = \log K$ instead of the exact root $b(t_n)$. For positive integer α the measure $\mathbb{P}_n$ is the size distribution obtained by choosing uniformly among the coloured objects; for every $t > 0$ it is exactly the conditional distribution of $\{k : X_k = 1\}$ given $S = n$ (source 48).

Corollary 10.1 (Gumbel law with explicit centring; sources 48 and 40). *Define*

$$\beta_\alpha(r) = (\alpha + 1)r - \log r + \alpha \log(\alpha + 1) - (\alpha + 1) \log \alpha. \quad (10.7)$$

For every fixed real x ,

$$\mathbb{P}_n \left(\frac{\alpha r}{K} L_n - \beta_\alpha(r) \leq x \right) \longrightarrow \exp(-e^{-x}). \quad (10.8)$$

For A022629 ($\alpha = 1$), an equivalent form is

$$\frac{L_n - \left(2K - \frac{K}{r} \log(r/2)\right)}{K/r} \implies G, \quad \mathbb{P}(G \leq x) = e^{-e^{-x}}. \quad (10.9)$$

Given Theorem 2.4, the corollary amounts to $t_n b(t_n) - \beta_\alpha(r) \rightarrow 0$: by (10.4), $t_n b = \alpha s b/M$ agrees with $(\alpha + 1)s - \log s + \alpha \log(\alpha + 1) - (\alpha + 1) \log \alpha$ up to $O_\alpha(\log s/s)$, and $s = r + O(r^{-2})$, $t_n = (\alpha r/K)(1 + O(r^{-2}))$ by (8.6) and (8.8). (The intake compared the two centrings numerically for $\alpha = 1, 2, 5$; the difference decays like $\log s/s$.) Source 48 proves it directly.

Proof (source 48). Work first with the unconditioned Bernoulli variables at the exact saddle, with s, M, t as in (2.7). Set

$$d = (\alpha + 1)s - \log s + \alpha \log(\alpha + 1) - (\alpha + 1) \log \alpha, \quad m = \left\lfloor \frac{d + x}{t} \right\rfloor.$$

Then $tm \rightarrow \infty$ and $m/M \rightarrow (\alpha + 1)/\alpha > 1$. Summation versus integration for a decreasing tail, followed by one integration by parts (or its standard real-power analogue), gives

$$\sum_{k>m} k^\alpha e^{-tk} \sim t^{-\alpha-1} (tm)^\alpha e^{-tm}. \quad (10.10)$$

For real $\alpha > 0$, the relative error from the integral asymptotic is $O_\alpha((tm)^{-1})$; the relative discretization error is $O_\alpha(t)$. Now $\log(1/t) = s - \log s - \log \alpha$ and $\log(d + x) = \log s + \log(\alpha + 1) + o(1)$. Hence

$$(\alpha + 1) \log(1/t) + \alpha \log(tm) - tm = -x + o(1),$$

and the tail in (10.10) tends to e^{-x} . The maximum individual odds on this tail are $O_\alpha(t)$ and tend to zero, so replacing odds by p_k changes the sum by $o(1)$, and $\sum_{k>m} p_k^2 = o(1)$. Independence gives

$$\mathbb{P}_t(X_k = 0 \text{ for every } k > m) = \prod_{k>m} (1 - p_k) \longrightarrow e^{-e^{-x}}.$$

It remains to justify conditioning on total size, which cannot be omitted. Let $T = \sum_{k>m} kX_k$ and $S^{\leq m} = S - T$. The same tail calculation with extra factors of k gives $\mathbb{E}_t T = O_\alpha(M)$ and $\text{Var}_t(T) = O_\alpha(M^2)$. By Lemma 4.1, $\mathbb{E}_t T/\sqrt{B} \rightarrow 0$ and $\text{Var}_t(T)/B \rightarrow 0$. The entire active interval near M lies below m for large n . Thus the shifted local limit estimate (5.8) applies to the truncated sum, and $\mathbb{P}_t(S^{\leq m} = n)/\mathbb{P}_t(S = n) \rightarrow 1$. Independence of the tail and the truncated sum now gives

$$\mathbb{P}_n(L_n \leq m) = \mathbb{P}_t(T = 0) \frac{\mathbb{P}_t(S^{\leq m} = n)}{\mathbb{P}_t(S = n)} \longrightarrow e^{-e^{-x}}.$$

This proves the result in the exact saddle coordinates. Finally, (8.6) and (8.8) imply $M/K = 1 + O_\alpha(r^{-2})$, $s = r + O_\alpha(r^{-2})$, and $t = (\alpha r/K)(1 + O_\alpha(r^{-2}))$. The change in the normalized extreme-value coordinate is $o(1)$, since L_n/M is tight by the limit just proved. Replacing s, M, t by $r, K, \alpha r/K$ therefore proves (10.8); tightness of the Gumbel coordinate also gives $L_n/K \rightarrow 1 + 1/\alpha$. For $\alpha = 1$, $\beta_1(r) = 2r - \log(r/2)$, which is (10.9). $\square$

The upper transition is near $M \sim K$, but the largest part is near $(1 + 1/\alpha)K$. There is no contradiction: a long tail of individually rare occupancies produces an extreme well beyond the transition (source 48).

Source 48 also proves the logistic part of Theorem 2.4 (its Proposition 7.2): for fixed distinct reals $u_1, \dots, u_d$ and $k_j = \lfloor M + u_j/t \rfloor$ at the exact saddle, the inclusion indicators converge jointly to independent Bernoulli variables with probabilities $(1 + e^{u_j})^{-1}$. Since $tH = s/(s - 1) \rightarrow 1$, its coordinate $u = t(k - M)$ and source 40's $x = (k - M)/H$ have the same limit, and its proof is the argument around (10.2): the log odds at k_j are $-u_j + o(1)$, and fixing finitely many indicators moves the remaining target by $O_\alpha(M)$ and removes variance $O_\alpha(M^2)$, both negligible.

11 A proved exponential scale from Fourier resonances

The bound (5.4) suggests an exponentially small scale not visible in either the logarithmic block or the central Edgeworth series. We can identify its constant at each fixed integer resonance.

Theorem 11.1 (Fixed-resonance modulus). *For every fixed nonzero integer ℓ , with ϕ_t as in Section 5,*

$$-\log \left| \phi_t \left(\frac{2\pi\ell}{M} \right) \right| \sim \frac{2\pi^4\ell^2}{3} \frac{H^3}{M^2} \sim \frac{2\pi^4\ell^2}{3\alpha^3} \frac{M}{s^3}. \quad (11.1)$$

Proof. The exact modulus formula is

$$\log |\phi_t(\theta)| = \frac{1}{2} \sum_{k \geq 1} \log \left(1 - 4p_k(1 - p_k) \sin^2(k\theta/2) \right).$$

At $\theta = 2\pi\ell/M$, write $k = M + xH$. For bounded x ,

$$\sin^2(\pi\ell k/M) = \pi^2\ell^2 x^2 H^2/M^2 + O_\ell(x^4 H^4/M^4).$$

The logarithm is therefore $-4p_k(1 - p_k)$ times this sine square to leading order. Lemma 4.1, applied with moments 2 and 4, gives

$$-\log |\phi_t(2\pi\ell/M)| \sim \frac{2\pi^2\ell^2 H^3}{M^2} \int_{\mathbb{R}} x^2 \frac{e^x}{(1 + e^x)^2} dx.$$

One may first restrict to $|x| \leq s^\delta$ with small fixed $\delta > 0$; Taylor remainders are then uniformly controlled. Outside this interval, the exponential boundary tails and the lower-endpoint bounds in Lemma 4.1 are negligible at the displayed scale. The quadratic logarithm error is $O(H^5/M^4) = o(H^3/M^2)$. Finally,

$$\int_{\mathbb{R}} x^2 \frac{e^x}{(1 + e^x)^2} dx = 4 \sum_{j \geq 1} \frac{(-1)^{j-1}}{j^2} = \frac{\pi^2}{3},$$

by expansion on the positive half-line. This proves the theorem. $\square$

For fixed α , the scale $\exp[-C_\alpha M/s^3]$ is smaller than every power of s/M , because M/s^3 dominates $\log M$. However, its constant deteriorates as α^{-3} . This makes the fixed- α asymptotic theorem compatible with large visible resonances at moderate n .

The theorem concerns the *modulus at a specified point*. It does not yet determine the shifted secondary saddles, their phases, or the contribution of their full neighborhoods to the coefficient integral. Calling it a complete multi-saddle transseries would be incorrect. It does identify a rigorous exponential scale and a concrete next target for exponential improvement.

12 Reproducible computation and finite-size diagnostics

12.1 Exact calculations

The supplied `code/40-norm-moments-verify.py` computes the five integer sequences through $n = 5000$ by descending polynomial multiplication. It compares their initial sixteen terms

against the consulted OEIS entries and independently checks $n \leq 150$ by a logarithmic-derivative recurrence. Explicitly, define

$$C_\alpha(m) = \sum_{d|m} (-1)^{m/d+1} d^{1+\alpha m/d}. \quad (12.1)$$

Then $a_\alpha(0) = 1$ and

$$na_\alpha(n) = \sum_{m=1}^n C_\alpha(m) a_\alpha(n-m). \quad (12.2)$$

This follows by expanding $qA'_\alpha(q)/A_\alpha(q)$ formally. The two algorithms have different arithmetic paths, so their agreement is a useful implementation check. It is not a proof of the asymptotic theorems.

All five exact prefix checks, all recurrence checks through 150, and strict monotonicity checks through 5000 passed. The complete computed lists are included as plain text. They were independently generated here, not copied from OEIS b-files.

12.2 The central approximation is excellent for one column, not all

Let $G = e^{F(t_n)+nt_n}/\sqrt{2\pi B}$, with the infinite sums evaluated by high-precision truncation. The following are numerical diagnostics, not interval-certified bounds. Values shown are rounded from the included CSV.

α	n	$a_\alpha(n)/G$	$a_\alpha(n)/(G(1+\mathcal{E}_1))$	$a_\alpha(n)/(G(1+\mathcal{E}_1+\mathcal{E}_2))$
1	100	0.989159702	0.999826713	0.999908591
1	1000	0.997526322	0.999995495	0.999999984
1	5000	0.999084078	0.999999367	0.999999999
2	5000	0.999545155	1.000036354	1.000036527
3	5000	0.961625508	0.961944240	0.961944315
4	5000	1.230518103	1.230825332	1.230825386
5	5000	1.581097698	1.581414178	1.581414223

At $\alpha = 1, n = 5000$, two central corrections reduce the relative error to about 1.1×10^{-9} . At $\alpha = 5, n = 5000$, the central approximation is plainly not accurate, and adding central corrections does not repair it. This is evidence about finite-size behavior, not a contradiction of a fixed-parameter limit. In particular, these tests must not be advertised as numerical confirmation of rapid convergence for all five columns.

The resonance diagnostic computes the exact product modulus at $\theta = 2\pi/M$, using the same numerical saddle:

α	n	$ \phi_t(2\pi/M) $
1	5000	2.33482×10^{-15}
2	5000	2.72621×10^{-5}
3	5000	0.0173352
4	5000	0.150645
5	5000	0.361805

The increase in these noncentral moduli is consistent with the observed failure of a central-only approximation. Their values do not by themselves predict the sign of the coefficient error; the omitted phases matter.

12.3 Logarithmic expansions are not monotone numerical improvements

For $\alpha = 1, n = 5000$, the exact logarithm is approximately 401.567349464797. Subtracting the leading expression $K(r - 1)$ leaves 41.05033086599. Subtracting the expansion through P_3 instead leaves -2.72431695876 , while truncating through P_7 leaves -7.67561845406 . The additional terms are not a promise of monotone improvement at a fixed small r . The missing prefactor and endpoint contributions are also on scales suppressed by the logarithmic theorem but relevant numerically.

An independent continuum check evaluates the integral and mean in Lemma 7.1, without using the rational-function expansion, and compares the normalized remainder after P_7 with P_8 :

α	s	Scaled remainder $r^8(\text{entropy}/(\alpha K) - \mathcal{H}_{\alpha,7}(r))$
1	24	7881.08213
1	48	5997.09948
1	96	5424.47363
Predicted limit $P_8(\pi^2/6)$:		4963.76905
3	24	15.7012553
3	48	9.85270737
3	96	9.49113831
Predicted limit $P_8(\pi^2/54)$:		9.15764483

The approach is slow, but has the expected direction in these samples. These are continuum diagnostics at nonintegral implied n , not exact integer-sequence computations at exponentially large indices.

12.4 What the numerical implementation guarantees

The integer arithmetic and recurrence comparisons are exact. For the saddle calculations, the program uses arbitrary-precision arithmetic and chooses a cutoff from a geometric majorant for $\sum k^{\alpha+6}e^{-tk}$. This bounds the tails relevant to all cumulants through order six, apart from harmless fixed constants. The test used 50 requested decimal digits with guard digits; the saved mean residuals are far smaller than the displayed precision. Nevertheless, the majorant and Newton iteration are evaluated in floating point, not outward-rounded interval arithmetic. Accordingly, the CSV values are diagnostics rather than certificates of rigorous numerical error bounds.

12.5 Source 48's computations for A022629 (source 48)

Two exact coefficient constructions. For an integer exponent $\alpha \geq 1$, multiply the product factors successively. Starting with $a_0 = 1$ and $a_n = 0$ for $n > 0$, process each k by updating in *decreasing* order of n : $a_n \leftarrow a_n + k^\alpha a_{n-k}$ ($n \geq k$). Descending order enforces at most one use of size k ; increasing order would compute a different partition problem. The independent check is the logarithmic-derivative recurrence (12.2). That formal identity is valid regardless of the analytic restriction in Proposition 3.4: only finitely many terms contribute to a fixed degree. The program `code/48-a022629-verify.py` compares the two constructions through degree 100, checks the initial OEIS values and strict monotonicity, and computes the whole exact table through $n = 6400$ for $\alpha = 1$ (`data/48-a022629-exact_coefficients.csv`).

[Merge note, batch 7301.] On the common range $n \leq 5000$ this table equals source 40's

`data/40-norm-moments-A022629_computed.txt` entry for entry; the intake also recomputed both from the product through $n = 6400$.

Saddle diagnostics. The exact saddle is found by Newton iteration, $t_{\text{new}} = t + (-F'(t) - n)/B(t)$, with a positivity safeguard. The first four cumulants use $b_2 = p(1 - p)$, $b_3 = p(1 - p)(1 - 2p)$ and $b_4 = p(1 - p)\{1 - 6p(1 - p)\}$. For $\beta \geq 0$ the explicit tail bound

$$\sum_{k > k_0} k^\beta e^{-tk} \leq e^t t^{-\beta-1} \Gamma(\beta + 1, t(k_0 + 1)) \quad (12.3)$$

follows by integrating $(u + 1)^\beta e^{-tu}$ over $u \geq k_0$. It controls truncation of the log product, mean, variance, and fixed-order cumulants after multiplication by a suitable polynomial constant. This addresses the infinite sums, but the implementation does not turn floating-point rounding into certified interval arithmetic.

n	$\log a(n)$	Gaussian relative error	Corrected relative error
100	31.57034412	$1.096 \cdot 10^{-2}$	$1.733 \cdot 10^{-4}$
400	80.97262500	$4.427 \cdot 10^{-3}$	$1.377 \cdot 10^{-5}$
1600	197.60929471	$1.849 \cdot 10^{-3}$	$2.527 \cdot 10^{-6}$
6400	467.20404426	$7.888 \cdot 10^{-4}$	$4.704 \cdot 10^{-7}$

Table 1: Computed diagnostics for A022629 ($\alpha = 1$), source 48. Relative error means approximation divided by the exact integer coefficient, minus one. The corrected approximation includes the two cumulant terms of (2.9). Calculations used 45 decimal digits.

At $n = 6400$, the bare logarithmic core underestimates $\log a(n)$ by about 45.36. Adding the first four inverse-logarithmic terms instead overestimates it by about 3.55. Thus that four-term logarithmic approximation is still poor as a *relative coefficient* approximation at this size, despite being a valid asymptotic expansion; the corrected exact-saddle formula has relative error approximately $4.7 \cdot 10^{-7}$. This is why Theorems 2.1 and 2.2 are stated separately. The recorded file `data/48-a022629-numerical_checks.json` gives the saddle, variance, cumulant correction, mean residual, cutoff, and tail estimate as well as the errors in Table 1. Its decimal values are computational evidence only; the exact integers and symbolic polynomial checks are exact computations. No Lean executable was run for either source.

[Merge note, batch 73O1.] The $n = 100$ row agrees with source 40's $\alpha = 1$ row in Section 12 ($1/0.989159702 - 1 \approx 1.096 \cdot 10^{-2}$, $1/0.999826713 - 1 \approx 1.733 \cdot 10^{-4}$), and source 48's saddle at $n = 100$ agrees with `data/40-norm-moments-run_alpha1.txt` to all printed digits.

13 Further research questions and proposed projects

The proved results leave several substantial and concrete directions.

1. An effective multi-saddle expansion. Locate the secondary saddles near $2\pi\ell/M$, compute their phases and Gaussian factors, and sum the relevant arcs with a uniform remainder. Theorem 11.1 supplies the first exponential action but not the full contribution. The moderate- n failures for $\alpha = 3, 4, 5$ make this a practical as well as theoretical priority.

2. A complete two-scale coefficient transseries. Refine the continuum-to-lattice passage sufficiently to retain endpoint terms, Euler–Maclaurin corrections, the central determinant, and

the resonance sectors in one consistent normalization. Determine which of these expansions are Gevrey and whether optimal truncation produces a useful exponential improvement. The present all-logarithmic series alone does not settle this.

3. The crossover as the tilt vanishes. The fixed- α expansion is singular as $\alpha \downarrow 0$, while the unweighted distinct-partition count has a different growth law. A natural crossover parameter is $\alpha \log K$. Derive a uniform family of saddle profiles when this parameter stays bounded, and match it to both the unweighted regime and the fixed-positive-tilt regime. The shrinking-tilt analysis in Gikunda’s thesis provides an important comparison point [10], but matching the full regimes still requires work.

4. Growing moments and concentration near maximal norm. Allow $\alpha = \alpha(n) \rightarrow \infty$. The fluctuation width H and resonance action show that several thresholds can occur before the model collapses onto maximal-norm partitions. Determine the sharp parameter ranges for a local limit theorem, for single-saddle dominance, and for localization near an extremal partition. The fixed- α proof does not justify substituting an arbitrary growing parameter.

5. The joint law of the largest several parts. The independent tail calculation suggests a Poisson process with intensity $e^{-x} dx$ for the rescaled upper parts. Prove its conditioned point-process limit and obtain joint distributions of the largest j parts, including second-order corrections to their centering. The maximum theorem is the one-dimensional starting point, not a substitute for tightness of the process.

6. Fluctuations of the number of parts and of the log norm. A multivariate conditioned local limit theorem should describe the number of occupied sites and the logarithm of the norm under fixed positive tilt. Conditioning removes a covariance direction and can cancel a leading variance term; computing the surviving scale requires more than plugging the unconditioned variance into a Gaussian formula.

7. More general weights. Replace k^α by $k^\alpha (\log k)^\beta$ away from the harmless small indices, or by a smooth regularly varying weight. Identify hypotheses that preserve a single large crossing, the logistic boundary, and a computable inverse expansion. Arithmetic thinning of the allowed part sizes would also require a replacement for the consecutive-block lemma and may introduce persistent lattice obstructions.

8. Certified numerics and formal verification. Give explicit constants in the Fourier bounds, add interval-enclosed saddle solving, and certify a chosen relative-error target at a stated n and α . For Lean, a sensible modular route is: the formal Euler product and recurrence; the Bernoulli coefficient identity; the consecutive-block inequality; an abstract triangular-array local-limit theorem; and finally the boundary smoothing lemma. The algebraic generation of P_j and the inverse coefficients can be separately certified by finite polynomial identities. This would connect directly to the repository’s transseries infrastructure without pretending that formal algebra alone proves analytic asymptotics.

13.1 Source 48’s proposed projects (source 48)

Source 48 lists eight next steps, stated for A022629 and the real family. They are proposed research tasks, not claims that the corresponding problems are all globally open. Its numbers

are kept; overlaps with the projects above are marked.

- 48.1 Resolve all subdominant asymptotic blocks** (overlaps project 2). The upper-transition calculation determines every K/r^j term. Determine the smaller contributions from the lower endpoint and the sum-integral discrepancy, including any powers of K , logarithms, and constants that appear for different α . Prove a fully explicit multiplicative equivalent with no implicit exact saddle sums. Distinguish this task from merely computing more terms in (2.2).
- 48.2 The transition as $\alpha \downarrow 0$** (overlaps project 3). At $\alpha = 0$ the model is ordinary distinct-part enumeration; for every fixed $\alpha > 0$ the leading logarithm has the different scale $\alpha K \log K$. Develop uniform asymptotics in a double limit, for example when $\alpha \log n$ stays bounded. The present proof explicitly does not cover this regime, and the change in the active interval makes it a useful test case for uniform asymptotic methods.
- 48.3 The number of parts after conditioning** (overlaps project 6). The finite-dimensional occupation profile is simple, but conditioning on total size creates long-range correlations. Determine the mean, variance, and limiting law of the total number of parts to several orders. A naive unconditioned variance calculation is not sufficient because number and size are strongly correlated in the narrow transition region.
- 48.4 Joint extremes and point processes** (overlaps project 5). Upgrade Corollary 10.1 to joint laws of the largest several parts and convergence of the rescaled tail point process. Prove quantitative error bounds, uniform tail estimates, and asymptotics of the expectation and variance of the largest part. Distributional convergence alone does not justify taking expectations.
- 48.5 Stable weight perturbations** (overlaps project 7). Replace k^α by $k^\alpha(\log k)^\beta$ or by $k^\alpha \exp(c_1/k + c_2/k^2 + \dots)$, with a positive convention for small k . Determine which coefficients and extreme-value centring terms are universal and which remember the perturbation. Establish explicit uniform hypotheses rather than assuming regular variation is automatically enough.
- 48.6 Optimal saddle remainder scales.** The absolute cumulant bound gives the safe parameter s/M . Signed cumulant cancellations can make actual errors much smaller, as Table 1 suggests. Determine the sharp first relative correction, its complete expansion, and the optimal error after each correction. This would improve both numerical evaluation and rigorous inverse brackets.
- 48.7 Effective inversion and formalization** (overlaps project 8). Turn the asymptotic O constants into explicit bounds, combine them with finite exact verification, and produce a certified algorithm for $N_\alpha(Y)$. A suitable ProveIt formalization can separate finite coefficient algebra, the positive product measure, the block minor-arc estimate, remainder transport, and integer-threshold bracketing. None of those formalization tasks should be replaced by unchecked numerical fits.
- 48.8 OEIS and literature follow-through.** Submit a concise, independently reviewed proof of (1.3) and the first sharper correction to the OEIS editors. Continue the literature audit for products with coefficients *inside* their factors, rather than mistaking $\prod(1+k^\alpha q^k)$ for $\prod(1+q^k)^{k^\alpha}$. Compare any earlier theorem at the level of hypotheses and error terms before making a priority claim.

14 Conclusion

The five checked OEIS conjectures share a single analytic mechanism. Filling parts up to a moving boundary produces the dominant factorial-scale entropy; smoothing that boundary produces the π^2 correction and its all-orders inverse-logarithmic successors. A conditioned Bernoulli model supplies the local probability that converts this entropy into an actual coefficient, and also gives logistic occupation and a Gumbel extreme. Lambert- W inversion turns the same expansion into a quantitative threshold law.

The resulting package contains explicit proofs rather than numerical extrapolation. It also records an important limitation: central asymptotics can converge extremely slowly for higher moments because noncentral resonances are only suppressed on the scale $\exp[-C_\alpha K/(\log K)^3]$. That proved scale and the visible finite-size discrepancies identify a well-defined next research problem: a phase-sensitive, exponentially improved multi-saddle expansion.

Source 48 reaches the same conclusions for A022629 by partly different routes: an elementary proof with a stronger additive remainder, an explicit differential operator that generates the entire dominant logarithmic block by finite exact algebra, a Lambert-based inverse for the integer threshold, a precise comparison with the largest individual partition weight, and a Gumbel law centred explicitly in $\log \sqrt{2n}$. The remaining distinctions—formal versus convergent series, logarithmic versus relative accuracy, and proved mathematics versus formal-machine verification—are essential parts of the result rather than editorial qualifications (source 48).

A Reproduction guide and file manifest

Besides this source, its compiled PDF and the README, the report directory contains the two sources' programs and recorded outputs under prefixed names.

File	Purpose
<code>code/40-norm-moments-verify.py</code>	Exact integer sequences, independent recurrence, and numerical saddle/Edgeworth diagnostics (source 40).
<code>code/40-norm-moments-derive_series.py</code>	Symbolic generation of the boundary derivatives and logarithmic coefficients.
<code>code/40-norm-moments-derive_inverse.py</code>	Symbolic reversion through six inverse-logarithmic orders.
<code>code/40-norm-moments-check_continuum.py</code>	Independent continuum quadrature diagnostics.
<code>code/40-norm-moments-check_resonances.py</code>	Direct numerical product moduli at the first resonance.
<code>data/40-norm-moments-*</code>	Computed integer lists, CSV diagnostics, symbolic output, execution logs, and the requirements file.
<code>code/48-a022629-coefficients.py</code>	Source 48's operator construction of six forward and six inverse coefficients.
<code>code/48-a022629-verify.py</code>	Source 48's exact table through $n = 6400$ and its saddle diagnostics.
<code>code/48-a022629-Makefile</code>	Source 48's build and verify targets (delivery layout).
<code>data/48-a022629-*</code>	Source 48's exact table, recorded checks, symbolic output, table fragment, and requirements.
<code>48-a022629-PROVENANCE.md</code>	Source 48's sources and verification record.

The mathematical proofs do not require running the programs. The programs write their

outputs under the names of the delivered packages, so they should be rerun on a copy of the delivered layout; the README gives the commands. The source uses standard LaTeX packages and no external bibliography processor. No Lean code or machine-checked proof is bundled or implied.

[Merge note, batch 7301.] Source 40’s manifest listed its files under their delivered names (`verify.py`, `derive_series.py`, `data/`, `README.md`, `requirements.txt`) with the commands `python verify.py --max-n 5000 --dps 50`, `python derive_series.py`, `python derive_inverse.py`, `python check_continuum.py`, `python check_resonances.py` and two `pdflatex` passes; source 48’s were `python code/coefficients.py --order 6` and `python code/verify.py --max-n 6400 --dps 45`. The shipped names are those above; the delivered READMEs are replaced by the report README.

B Compact proposed OEIS formula updates

For the entry corresponding to $\alpha = 1, \dots, 5$, write $K = \sqrt{2n}$ and $r = \log K$. The following common formula is proved in Theorem 2.1:

$$\log a_\alpha(n) = \alpha K(r - 1) + \frac{\pi^2 K}{6\alpha r} + \frac{\pi^2 K}{6\alpha r^2} + O_\alpha(K/r^3).$$

It is appropriate to link a reviewed proof before changing a database’s conjectural status. This report has not submitted or edited any OEIS entry. For the array [A292189](#), the result holds down every fixed positive column; it makes no uniform assertion about the main diagonal or columns whose index grows with the row.

Source 48’s suggested A022629 addition

Source 48 suggests the following text for independent editorial review; it has not been submitted.

The logarithmic conjecture follows with the stronger estimate $\log a(n) = \sqrt{2n}(\log \sqrt{2n} - 1) + O(\sqrt{n}/\log n)$. More precisely, with $K = \sqrt{2n}$ and $r = \log K$,

$$\log a(n) = K \left(r - 1 + \frac{\pi^2}{6r} + \frac{\pi^2}{6r^2} + \frac{\pi^2/6 - \pi^4/72}{r^3} + O(r^{-4}) \right).$$

A proof, an all-orders coefficient rule, and a separate relative saddle-point formula are supplied in the accompanying research report.

A particularly short proof of the first assertion uses the partition $(1, 2, \dots, m-1, m+d)$ with $n = m(m+1)/2 + d$ for the lower bound, and $a(n) \leq e^{tn} \prod_k (1 + ke^{-tk})$ with $t = \log \sqrt{2n}/\sqrt{2n}$ for the upper bound. Splitting the logarithm into its positive-part contribution and a smoothing remainder of width $O(\sqrt{n}/\log n)$ finishes the argument; see Section 3.1.

[Merge note, batch 7301.] The “accompanying research report” is now this report.

C Verification boundaries and source audit of source 48

Item	Boundary of the claim
OEIS status	A022629 was retrieved with the 2018 formula still labeled a conjecture. This is evidence about that entry, not a guarantee of absence of a proof elsewhere.
Repository inspection	The repository README and the Transseries and Inversion README were read; a targeted A022629 code search returned no match. No complete audit of every research artifact is claimed.
Mathematical results	Conventional proofs are given in the body of this article. The proof of the displayed OEIS conjecture is independent of the more elaborate local limit and formal-expansion arguments.
Symbolic calculations	Six forward and six inverse coefficients are generated by exact rational-polynomial arithmetic in c using the stated operator recurrence.
Integer computations	The product recurrence is evaluated through $n = 6400$; the independent logarithmic-derivative recurrence is checked through $n = 100$.
Floating-point diagnostics	The saddle comparisons use 45 decimal digits and explicit analytic truncation bounds. They are not interval-certified evaluations.
Formal verification	No Lean or Rocq/Coq proof of the new analytic theorems is included or claimed. Repository formalization material is a possible integration path, not a certificate of this article.
Novelty	No worldwide priority or “breakthrough” status is certified. The contribution is the supplied proof and its stated extensions.

D Provenance and merge record

Source	Archive (arrival commit)	Main file, PDF	Pin	Author line
40 (base)	OEIS_Distinct_Partition_Norms.zip (f8c3a392a)	article.tex, 22 pp.	1f1981f68	Research report prepared for Vladimir Reshetnikov
48	A022629_Research_Package.zip (3a9518c52)	article.tex, 20 pp.	dc1a7242d	Research report prepared for Vladimir Reshetnikov; developed with AI assistance; proofs and computations supplied for review

Both are manuscripts of batch 73 (cluster O1) of ProveIt’s incoming-reports intake, numbered 40 and 48 there; the placement commit [9df4ba51a](#) staged their files and deleted the archives, which survive in their arrival commits. Source 40 arrived first. Neither cites the other, and their texts share almost nothing (8-word shingle containment 0.004 both ways; the longest common run is a `pdflatex` command line): they are independent. Both pins are ProveIt commits of 1 October 2026; source 48 calls its identifier a “tree response” and does not assert that it pins every subsequent read.

What each contributed. Source 40: the base text, in its order, and everything in it, including the five sequences, P_7 , P_8 , the resonance theorem and the finite-size analysis. Source 48: the second proofs of Proposition [3.2](#), Lemmas [4.1](#) and [5.1](#), Theorem [2.2](#) and the threshold transfer;

Propositions 3.4 and 3.5; Lemma 3.6; the operator route; Corollaries 8.1 and 10.1; the A022629 forms; its computations, projects, OEIS wording and audit table.

Where the merge chose.

- Base: source 40, the more general manuscript (five sequences, one more coefficient, the resonance theorem).
- Notation: source 40's throughout; source 48's symbols renamed as in Section 1.5.
- Source 48's λ -centred Gumbel law is printed as a corollary of Theorem 2.4 with its own proof, since its centring is explicit where source 40's is the exact root $b(t_n)$.
- Source 48's statements that duplicate source 40's (its Theorem 1.1, Theorem 2.4, Lemma 3.1, Lemma 3.2, Theorem 3.3, Theorem 6.1, Proposition 7.2) are printed once, credited, with only their different proofs or forms kept. Its abstract, title page and package-contents paragraph are replaced by this text and the README.
- Both data tables of A022629 are shipped (they agree on $n \leq 5000$).
- The two bibliographies are merged; source 48's OEIS item is source 40's [1].

References

- [1] OEIS Foundation Inc., *A022629: Expansion of $\prod_{m \geq 1} (1 + mq^m)$* . Conjectural formula attributed to V. Kotešovec, May 8, 2018. Internal entry consulted October 1, 2026; sequence revision #54, March 13, 2026. <https://oeis.org/A022629/internal>.
- [2] OEIS Foundation Inc., *A092484: Expansion of $\prod_{m \geq 1} (1 + m^2 q^m)$* . Conjectural formula attributed to V. Kotešovec, December 27, 2020. Consulted October 1, 2026. <https://oeis.org/A092484>.
- [3] OEIS Foundation Inc., *A265840: Expansion of $\prod_{k \geq 1} (1 + k^3 q^k)$* . Conjectural formula attributed to V. Kotešovec, December 27, 2020. Consulted October 1, 2026. <https://oeis.org/A265840>.
- [4] OEIS Foundation Inc., *A265841: Expansion of $\prod_{k \geq 1} (1 + k^4 q^k)$* . Conjectural formula attributed to V. Kotešovec, December 27, 2020. Consulted October 1, 2026. <https://oeis.org/A265841>.
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